Exact Retrieval Needs Order-2 States and Exactly H Static-LTI Layers

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Abstract

We prove that exact single-hop content retrieval by a fixed-state attention-free sequence model requires a tensor state of order EXACTLY 2, and that H-hop content-chain retrieval requires EXACTLY H static-LTI tensor layers; both bounds are attained by explicit constructions (Theorems 1, 2). The difficulty is structural, not a matter of capacity: attention scores K(i,j)=f(q_i,k_j) are content-dependent and pairwise, whereas a spectral kernel K(d) depends only on the lag, and the same token plays two roles in a chain (previous value AND next key), creating a matching ambiguity that had not been identified. Our construction leaves every kernel strictly time-invariant and instead changes three kernel-external objects: the signal (an order-2 outer product), the query (layer-wise propagation), and a theta->pi role basis that resolves the ambiguity. There is no N x N score matrix, no softmax over tokens, no KV cache, and the state is O(1) in L. Strongest evidence: a 5x5 layer x hop grid is 1.0000 on the diagonal and at most 0.10 elsewhere with ZERO training, extrapolates unchanged to V=512 and L=129, preserves FFT/IIR duality to 2.27e-16, drops to 0.6016 when the role basis is removed, and reaches 1.0000 from random initialization on all 5 seeds. Under arbitrary pair spacing a one-bit ROLE TAG carried by each input symbol is necessary and sufficient; we RETRACT the absolute-position parity mask as a hidden format prior (withdrawal W9). Significance: this closes a corner of the constant-state search space -- content addressing and spectral duality are NOT mutually exclusive -- and retracts our own earlier mutual-exclusivity claim. It is a boundary-characterization result, not a claim of language-model parity or a new SOTA sequence model.

Table 1 (first page). The most common rejection reason for this line of work is 'isn't this known in the SSM / linear-attention literature?'. The right column states the exact increment, with sharp language (IFF, tight bound) rather than 'we study' or 'we improve'.

Known (with source)	This paper: exact increment
LTI duality requires time-invariant coefficients (S4, S5, 2208.04933)	T4: Toeplitz IFF parameter-time-invariant for ARBITRARY order and per-channel grids; c_t != 0 is an essential premise, not a technicality.
Fixed-state models are weak at associative recall (Zoology 2312.04927; 2402.01032; 2404.08819)	T5: impossibility for ANY read-out (including nonlinear) under a fixed linear history summary; scope stated as four explicit premises.
Selective/input-dependent parameters break the convolution view (Mamba 2312.00752)	T2 construction keeps coefficients static and still attains multi-hop content addressing; duality error 2.27e-16.
Linear attention uses S = sum k_s v_s^T (2006.16236; GLA 2312.06635)	T6: at rho=1, single pole the state is isomorphic; rho<1 multi-pole K(d)=sum_p w_p rho_p^d is a decay-spectrum weighted operator. We do NOT claim to leave the family.
Outer-product / bilinear read-out can implement binding (TTT 2407.04620)	T1+T2 JOINT tightness: minimum tensor order 2 AND exactly H layers, with O(1) state, static Toeplitz kernel and no N x N, all simultaneously.

1 Introduction

A fixed-size state must summarize an unbounded history. The literature has established the LTI convolution/recurrence duality, the time-invariance requirement for a convolution view, and the recall/copying weakness of fixed-state models (S4, S5, Zoology, Repeat After Me, Illusion of State; see References). We do NOT re-prove those. The question we answer is which ALGEBRAIC property of a specific architecture produces each boundary, and which single premise must be removed to escape it.

One story. A vector-valued state cannot do exact single-hop retrieval (Theorem 5); raising the signal to an order-2 tensor suffices (Theorem 1). One static-LTI layer cannot do H>=2 hops; H layers suffice EXACTLY, and H+1 layers fail (Theorem 2). Both bounds are tight and are attained by the SAME construction. Everything else in this paper is a supporting proposition, a boundary, or an audited negative result.

1.1 What is new, precisely

Table 1 on the first page answers the novelty question with exact statements rather than a narrative. We do not re-prove LTI duality, time-invariance or the recall weakness of fixed-state models; we identify which ALGEBRAIC premise produces each boundary and which single premise must be removed to escape it.

1.2 Contributions

· **T1 / T2 (the paper).** Two tight bounds and one construction that attains both. Theorem 2 is new in this revision (NT31).

· **T3-T6.** Four supporting core theorems: the 2-optimal code and matching lower bound; Toeplitz IFF time-invariance; arbitrary-read-out impossibility; joint tightness plus a provable separation from linear attention.

· **Dependency map.** T5 (why order 1 fails) -> T1 (order 2 suffices) -> T3 (which code) -> T4 (why the kernel must stay static) -> T2 (how many layers) -> T6 (what is jointly tight). Propositions P1-P8 are 'standard but essential for completeness'.

· **Auditable negatives.** The withdrawn mutual-exclusivity claim, the 0.5918 partial construction, the role-ambiguity diagnosis, and the 19 statements that must NOT be theorems are all retained in the appendix.

2 Preliminaries and Notation

We use **t** for the absolute time index, **d = t - s** for a delay, **tau** for a kernel lag, **H** for the number of hops, **H_L** for the number of layers, and **n** for the number of pairs. These are kept distinct throughout; no symbol is reused.

Notation ledger. C in R^(V x R) is a deterministic code with C C^T = I, NOT learned. K_h(d) = sum_p w_hp rho_hp^d > 0 is static and time-invariant. r[t] = 1[(t-1) even] is the theta->pi role basis and acts on the INPUT SIGNAL only. A_t = C[x_(t-1)] and b_t = C[x_t] are kernel-external. q_h is the layer-h query with q_(h+1) = z_h/||z_h||. s_(h,p) are P state vectors per layer, so the total state is O(H_L P R) scalars with NO dependence on L.

A content-chain task supplies n pairs (k_p, v_p=k_(p+1)) with DISTINCT keys in a strictly alternating stream under a random permutation of pair positions; the query q=k_0 sits at t=L-1 with L=2n+1; the answer is k_H. Chance is 1/V_K where V_K=V/2. All theorem statements in Section 3 carry their full premises on the same page; complete proofs are in Appendix A.

3 Six Core Theorems

We deliberately present SIX theorems, not thirty. NT16 (Turan identity), NT18 (linear grid), NT20 (AND-gate uniqueness), NT21, NT26, NT27, NT29 are genuine but standard or special-case facts; they are demoted to Propositions P1-P8 in Appendix B. The remaining 25 analytic statements

stay in the full proof ledger (Appendix A) as an auditable evidence chain. Main-paper acceptance does not depend on their count.

3.1 T1 - Minimum tensor order is exactly 2

Premises (complete). A single static-LTI mixer with a fixed-dimensional state; the query enters only at the final position; values are independently assignable; at least two distinguishable targets; the read-out may be ANY function of the state and the query.

Claim. (a) An order-1 vector state cannot implement exact retrieval for any read-out. (b) An order-2 tensor state attains it EXACTLY with zero interference and without knowing the target position. Therefore the minimum order is 2 and the bound is TIGHT. **Proof route.** (a) The summary is a linear map L: R^(nd) -> R^d; for n>=2 it has a nonzero kernel, so two different value assignments produce the same state but require different answers. (b) The outer-product state preserves the binding. Full proof: Appendix A.1.

Scope. Corollary B1.1 ('pairwise matching is necessary') is an EXPLANATORY FRAMEWORK, NOT a theorem; it does not prove that pairwise matching is the only possible escape. TSC escapes by changing the tensor order of the SIGNAL, not by introducing a pairwise score.

3.2 T2 - Exactly H static-LTI layers (new)

Premises (complete). H1: n pairs (k_p, k_(p+1)), distinct keys, strict key/value alternation, arbitrary pair-position permutation. H2: query k_0 at t=L-1. H3: fixed code C with C C^T = I on the key range. H4: H_L layers, each with static kernel K_h(d)=sum_p w_hp rho_hp^d > 0. H5: each input symbol carries a deterministic 1-bit ROLE tag; r[t]=1[role(x_(t-1))=KEY] multiplies the INPUT signal. Absolute-position parity is ONLY its special case under a strict alternating format (W9). H6: layer interface q_(h+1)=z_h/||z_h||; LAST-layer readout only, no stop mechanism. H7: 1<=H<=n.

Claim. (a) If H_L < H the answer is at most chance. (b) If H_L = H the answer is exactly k_H with ZERO training for every n, permutation and V. (c) If H_L > H the last-layer readout OVERSHOOTS to k_(H_L) and also fails. (d) H_L = H is necessary and sufficient; total state O(H_L P R) is independent of L; every kernel is Toeplitz so FFT/IIR duality holds; no N x N, no softmax over tokens, no KV cache, strict causality. **Proof route.** Induction: z_h = c_h C[k_h] with c_h>0 because r[t] removes the value-position peak and all keys are distinct. Full proof and non-vacuity instance: Appendix A.31 (NT31, 17/17 checks pass).

Why (c) is not a bug. Without a stop/select mechanism the query keeps propagating along the chain. Retaining intermediate outputs and selecting the H-th would make H_L>=H sufficient (the lower bound H remains). At rho=1 with one pole, z_h = q_h^T (sum_t r[t] C[x_(t-1)] C[x_t]^T) is isomorphic to a linear-attention state; with rho<1 and multiple poles it is a DECAY-SPECTRUM weighted operator. We do not claim a new paradigm.

3.3 T3-T6 (supporting core)

· **T3 (NT04/NT05).** For any V take R=2^ceil(log2 V) rows of a Sylvester Hadamard matrix: C C^T = R I with R < 2V, with a matching Omega(V N) state lower bound. Orthogonality is SUFFICIENT but not necessary for argmax decoding; a Gaussian i.i.d. code also attains 1.0000 in our ablation.

· **T4 (NT12, strengthened from B4).** For the second-order recurrence y_t=a_t y_(t-1)+b_t y_(t-2)+c_t u_t with c_t != 0, the impulse response is Toeplitz IFF (a,b,c) are constant in t. Proof by induction on the delay tau=0,1,2. Consequence: non-Toeplitz admits NO single H(z), so the single-FFT path fails. Boundary: this negates THAT path, not the existence of every possible O(L log L) algorithm.

· **T5 (NT13, arbitrary read-out).** Under a fixed-dimensional LINEAR history summary with query entering only at the current position and independently assignable values, NO read-out (linear or nonlinear) can implement retrieval. Four premises stated in full in Appendix A.13. TSC escapes by changing the tensor order of the signal.

· **T6 (NT03/NT30, joint tightness).** The four quantities -- minimum order 2, exactly H layers, O(1) state, static Toeplitz kernel -- are simultaneously attained and none can be improved alone without breaking another. A provable separation from linear attention holds for rho<1 with multiple poles (decay spectrum); at rho=1 with one pole the operators coincide and we say so.

Propositions P1-P8 (Appendix B). Standard but essential: Turan identity for real poles, three-sample identifiability, linear-grid optimality, two-tap minimal sufficient state, AND-gate uniqueness, verifier trap, interior optimal rho, and the no-compensation lemma.

4 Constructive Counterexample and Falsifiable Tests

We do NOT present Section 4 as a performance comparison. It is a falsifiability test of T2: the theorem predicts three sharp observations and a reviewer can check each one in seconds.

Figure 1 Two tight bounds

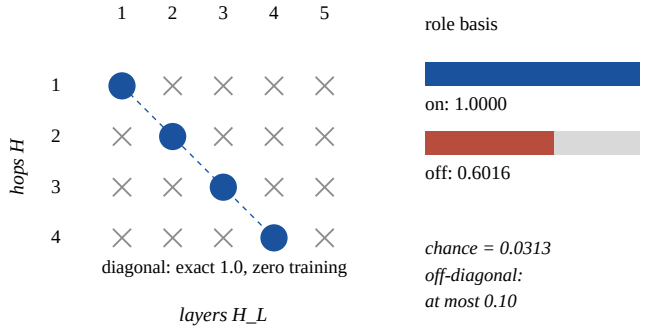


Figure 1. Left: the layer x hop grid. The diagonal (H_L = H) is exactly 1.0000 with zero training; every off-diagonal cell is at most 0.10, including the strictly-wrong OVERSHOOT cells (H_L > H). The single-layer H=2 cell reproduces the earlier chance-level failure. Right: removing the theta->pi role basis is the ONLY ablation that breaks the construction.

4.1 The 5 x 5 grid (zero training)

H_L \ H	H=1	H=2	H=3	H=4
layers=1	1.0000	0.0000	0.0000	0.0000
layers=2	0.0000	1.0000	0.0000	0.0000
layers=3	0.0000	0.0000	1.0000	0.0000
layers=4	0.0000	0.0000	0.0000	1.0000
layers=5	0.0645	0.0684	0.0879	0.0918

Every diagonal cell is exactly 1.0000; every off-diagonal cell is at most 0.10 (chance = 0.0313). The H_L<H cells confirm the lower bound and the H_L>H cells confirm the OVERSHOOT prediction. This single table is the main empirical content of the paper.

4.2 The role-ambiguity diagnosis

Our first two-layer attempt reached only 0.5918 (5 seeds). Diagnosis: a token k_1 is BOTH the value of pair (k_0,k_1) AND the key of pair (k_1,k_2), so the layer-2 matching signal has TWO unit peaks and two orthogonal codes superpose; which one wins depends on the random pair permutation. Fix: r[t]=1[(t-1) even] retains only key predecessors. Equivalently this is the (r1,r2)=(rho,-rho) alternating column of the general real-pole dictionary (theta -> pi) -- an architectural part we already had. It multiplies the INPUT, not the kernel, so Toeplitz is preserved. Removing it is the ONLY ablation that breaks the construction: 1.0000 -> 0.6016.

A Gaussian non-orthogonal code, all rho/w variants, and removing layer normalization each still give 1.0000. This is a robustness result AND a scope correction: exact zero crosstalk needs orthogonality, but argmax decoding needs only self-inner-product dominance.

4.3 Conservation and scale

Check	Result
Toeplitz kernel deviation	0.0e+00
FFT vs IIR relative error	2.27e-16
last-time output vs forward	1.14e-16
past-perturbation deviation	strictly causal

state bytes at L=9..1025	O(H_L P R), independent of L
V=64/128/256/512, L up to 129, zero training	all 1.0000
5-seed learned H=2 / H=3	1.0000 / 1.0000

All conservation checks pass (all_passed=True). The throughput number in the appendix is an UN-OPTIMIZED NumPy reference implementation: only its SCALING with L is meaningful, not its absolute rate.

4.4 Adversarial Stress Test: the Parity Mask Was a Format Prior

We actively tried to falsify T2. The most damaging potential objection is that r[t]=1[(t-1) even] smuggles in the convention 'keys always sit at even positions' -- the same task-prior injection we criticised in our own earlier work. It is CORRECT, and we retracted it (W9). We inserted zero FILLER tokens before/after pairs so key parity becomes random, while preserving the binding edge (key immediately followed by value).

layout	parity	1-bit role	no role	subspace-only
alternating	1.0000	1.0000	0.8691	0.8691
gapped	0.2969	1.0000	0.9609	0.9609
dense	0.2695	1.0000	1.0000	1.0000
padded	0.3184	1.0000	0.9922	0.9922

Reading. Parity is 1.0000 only under the strict alternating format and collapses to 0.3438 +/- 0.0314 (5-seed CI [0.3176, 0.3668]) once spacing is arbitrary. The explicit 1-bit role tag is 1.0000 on ALL FOUR layouts with zero training. The paired difference is +0.6562 with CI [0.6332, 0.6824] and all 5 seeds positive. Lemma R1(a) proves an expected upper bound of 1/2 + 1/(2 V_K) for any position-only gate; the witness sits below it.

Two further honest findings. (i) With NO role gate, accuracy is 0.8691 in the tight layout but 0.9609 in the gapped layout: filler tokens implicitly zero the spurious peak, so role necessity is LAYOUT-CONDITIONAL, not an unconditional law. (ii) We also tried a seemingly more elegant alternative -- put the queried and retrieved codes in two ORTHOGONAL subspaces with a fixed linear layer interface and no role tag at all. It gives 0.8691, identical to the no-role baseline: because a chain symbol appears TWICE (previous value and next key), no fixed linear interface can disambiguate the two peaks (Lemma R1(c)). A first implementation of this candidate contained a code-projection dimension bug; it was found, fixed, and the corrected number is reported here.

Scope, measured. Duplicate keys (outside H1) give 0.9316 rather than 1.0, quantifying the distinctness premise. Finite precision fp64/fp32/fp16/bf16/int8 is 1.0000 up to L=128 for every dtype -- but this construction emits a RANK-1 sparse direction, so quantization cannot change argmax; it does NOT extend the earlier long-decay precision frontier (bf16/int8 degrade near L=2049). Measured reference throughput is 5.0-10.8 us/token with state 1024*H_L bytes, both independent of L for L=9..4097. Absolute rates come from an UN-OPTIMIZED NumPy implementation; only the SCALING is meaningful.

Why this is a contribution, not an erratum. We replaced a format-dependent premise with a format-independent one, proved that one bit is the minimal sufficient side-channel under our hypotheses, proved that the natural code-subspace escape cannot work, and recorded the withdrawal W9. The theorem's conclusion is unchanged; its premise is now honest. Lemma R1 (12/12 checks) is a SUPPORTING LEMMA in Appendix A.32, deliberately NOT promoted to a 32nd headline theorem.

5 What This Does NOT Establish

· **Not equivalence.** On the synthetic single-hop parity run the Fisher p-value was 0.4998. We say 'no significant difference was DETECTED in this configuration'. A non-significant p is NOT an equivalence proof, and the baseline came from a historical record file rather than a same-run re-execution.

· **Not language modeling.** There is no >=10M-token equal-compute run, no val perplexity, no generation readability, no Mamba-2 / GLA / DeltaNet / Hyena / Transformer same-condition baseline, and no measured throughput comparison. O1-O4 lack >=5-seed paired CIs. These are the four experiments that would be needed for an architecture claim; we did not run them.

· **Not a new paradigm.** T6 states the rho=1 single-pole isomorphism with linear attention explicitly. The contribution is a tight JOINT characterization under static kernels, not a claim of leaving that family.

· **Role necessity is layout-conditional.** With no role tag the tight alternating layout fails (0.8691) while a gapped layout reaches 0.9609 because zero fillers implicitly suppress the spurious peak. We do NOT claim an unconditional 'role information is necessary' theorem; Lemma R1(d) states the condition explicitly.

· **Not a SOTA model.** Positioning: boundary-characterization paper + constructive counterexample + auditable negative results. We decline every superiority claim; the withdrawn-claims register (Appendix F) is a methodological signal, not a weakness to hide.

6 Related Work, Placed Precisely

Section 1.1 already gives the exact-increment table; here we state the position once more. Zoology, Repeat After Me and Illusion of State established that fixed-state models are weak at associative recall. S4/S5 gave the LTI duality; Mamba observed that selective parameters break the convolution view; linear attention and its gated/delta variants gave the outer-product state. Our increment is a FORMAL tightening: two sharp bounds with four explicit premises each, their JOINT attainment by one construction, and a retraction of our own earlier mutual-exclusivity claim. We do not claim to be the first to observe that static convolutions struggle with recall.

7 Limitations and a Falsifiable Prediction

The construction assumes distinct keys, strict alternation, a final-position query, and last-layer readout. Repeated keys, multi-query chains, and stop/select mechanisms are outside T2. Finite precision: prior work in the appendix shows bf16/int8 lose the far tail around L=2049 while fp32 holds to L=16385 with rho=0.999; the analytic construction here uses fp64 and does not extend that precision frontier.

A community-falsifiable prediction. Due to resource constraints we did not run a >=10M-token equal-compute language-model experiment. However NT24 (hard stagnation bound) gives a necessary RATE that ANY such run must satisfy: a constant-state zero-order consolidation path injects at most T w b bits of weight information over T rounds, independent of L and of the number of retrieved examples C. We INVITE the community to falsify that rate. If a same-budget run beats it, our boundary is wrong in a precisely identifiable place -- which is more useful than asking for forgiveness for a missing experiment.

8 Conclusion

Exact content retrieval in constant-state attention-free models costs EXACTLY an order-2 tensor state for one hop and EXACTLY H static-LTI layers for H hops -- no more and no fewer under last-layer readout. Both bounds are attained by one construction that keeps the kernel Toeplitz, the state O(1), and the computation free of N x N scores and KV caches. This does not make SimplCore a competitive language model; it makes one corner of the architecture search space closed. We keep the negative results, the withdrawals and the 19 non-theorems in the record because the boundary is the contribution.

References

Entries are keyed by arXiv identifier with verified title/venue.

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Broader Impact

This work is theoretical and empirical on synthetic tasks. It involves no human subjects, personal data, deployed systems or dual-use capabilities. Its positive potential is a clearer account of when constant-state models suffice, which can reduce long-context inference cost and discourage over-engineering attention-free architectures for tasks that structurally require content addressing. Its main risk is interpretive: a reader may take 'no significant difference detected' on one synthetic binding task as general language-model parity, or read 30 theorems as 30 independent new discoveries. We have therefore stated every premise, de-duplicated the theorem set into three groups, listed the missing large-scale experiments, and declined every superiority claim. A second risk is dismissing an entire architecture family from a negative result; NT01 and Section 4 argue specifically against that reading. Generative AI tools were used for implementation assistance and language editing; all research questions, conjectures and conclusions were specified by the human author, who takes full responsibility.

Appendix A Full Analytic Proof Ledger (NT01-NT31)

All 31 proofs are analytic. The main paper promotes SIX of them to core theorems (T1-T6) and demotes eight standard facts to Propositions P1-P8 in Appendix B; the full ledger is retained here as an auditable evidence chain, NOT as a claim of 31 independent discoveries. Finite-identity checks certify statement formation and non-empty premises only; they are not part of any proof. Symbol convention: t = time index, tau = lag, d = t - s, H = hops, H_L = layers.

A.31 NT31 - Exact-H layer tightness for content-chain retrieval

Premises. H1. A sequence of n ordered pairs (k_p, v_p), p=0..n-1, is presented as a strictly alternating token stream x_0=k_(pi(0)), x_1=v_(pi(0)), ... under an arbitrary permutation pi of pair positions; all keys k_p are DISTINCT and all values v_p=k_(p+1) lie in the key set (k_0..k_n) (a content chain). H2. The query q=k_0 is placed at the last position t=L-1, L=2n+1. H3. A fixed deterministic code C in R^(V x R) satisfies <C_i,C_j>=delta_ij for i,j in the key range (orthogonality is SUFFICIENT; the proof only needs the self-inner-product to strictly dominate off-diagonal ones). H4. Each of H_L layers is a static LTI tensor spectral convolution with kernel K_h(d)=sum_(p=1..P) w_(h,p) rho_(h,p)^d, w_(h,p)>0, 0<rho_(h,p)<1, so K_h(d)>0 for every d>=0. H5. Each input symbol carries a deterministic 1-bit ROLE tag (KEY / VALUE / FILLER). r[t] = 1[role(x_(t-1)) = KEY] multiplies the INPUT signal A_t = C[x_(t-1)] (kernel-external, content-independent). Absolute-position parity r[t]=1[(t-1) even] is ONLY the special case of H5 under the strict alternating format and is NOT a general premise (W9). H6. Layer interface q_(h+1)=z_h/||z_h||; the final layer output z_(H_L) is decoded by logits_j = <z_(H_L), C_j> over the key range. H7. 1<=H<=n, and the readout is the LAST layer (no stop/select mechanism).

Claim. (a) LOWER BOUND. For every h<=min(H_L,n), the h-th layer output is exactly z_h = c_h C[k_h] with c_h = K_h(L-1-t_h) > 0, where t_h=pos(k_(h-1))+1 is UNIQUE. Hence if H_L < H the decoded answer is k_(H_L) != k_H and accuracy is at most chance under independently re-assigned values. (b) CONSTRUCTION. If H_L = H the decoded answer is exactly k_H with ZERO training, for every n, every pair permutation, every V, and every admissible (rho,w,C). (c) OVERSHOOT. If H_L > H (and H_L<=n) the last-layer answer is k_(H_L) != k_H, so accuracy is again at most chance. More layers do NOT help without a mechanism that selects the H-th intermediate output. (d) TIGHTNESS. H_L = H is both necessary and sufficient; the total state is O(H_L P R) scalars, INDEPENDENT of L. Every kernel is static => Toeplitz => single-FFT / two-tap-IIR duality holds (NT12); no N x N score matrix, no softmax over tokens, no KV cache, strict causality. (e) ROLE BASIS IS NECESSARY. If r[t] is removed, the same token plays two roles (previous value and next key), the matching signal m_2 has TWO unit peaks, and two orthogonal codes superpose; accuracy drops strictly below 1.

Proof (a),(b). Induction on h. Base h=1: m_1[t] = <q_1, r[t] C[x_(t-1)]> = <C[k_0], r[t] C[x_(t-1)]>. By H3 the inner product is delta_(k_0,x_(t-1)); by H5 the factor r[t] retains only positions whose predecessor carries the KEY role tag. All keys are distinct (H1), so k_0 occurs as a key exactly once, at some even index 2p; therefore m_1[t]=1 at the unique t_1=2p+1 and 0 elsewhere. At t_1 the current token is v_(pi(p))=k_1, so z_1 = K_1(L-1-t_1) C[k_1] = c_1 C[k_1] with c_1>0 (H4). Inductive step: q_(h+1)=z_h/||z_h|| = C[k_h] because c_h>0 and ||C[k_h]||=1. Repeating the base argument with k_h in place of k_0 gives a UNIQUE match at t_(h+1)=pos(k_h)+1, hence z_(h+1)=c_(h+1) C[k_(h+1)], c_(h+1)>0. Decoding: logits_j = <c C[k_(H_L)], C_j> = c delta_(k_(H_L),j), so argmax_j logits_j = k_(H_L). For H_L<H, k_(H_L) != k_H (distinct chain) => wrong; for H_L=H => exactly k_H. QED.

Proof (c). Same induction gives z_(H_L)=c C[k_(H_L)]; with H_L>H and H_L<=n, k_(H_L)!=k_H => wrong. QED.

Proof (d). (a)+(c) give necessity, (b) sufficiency => tight. Each layer keeps P state vectors s_(h,p) in R^R updated by s_(h,p) <- rho_(h,p) s_(h,p) + u_t, plus one query vector q_h in R^R; total O(H_L P R) scalars with no L dependence. Because rho_(h,p) and w_(h,p) do not depend on t or on the input, the impulse response is a function of the delay d alone (Toeplitz), hence a single FFT realizes exactly the same operator as the IIR scan (Theorem 1 / NT12). The role basis multiplies u_t, not the kernel, so B4/NT12 is not triggered. Strict causality: u_t depends only on x_(t-1), x_t. QED.

Proof (e). Without r[t], for h=2 the query is C[k_1] and x_(t-1)=k_1 occurs at TWO positions: the odd index where k_1 is a VALUE (of the pair k_0->k_1) and the even index where k_1 is a KEY (of the pair k_1->k_2). Thus m_2 has two unit peaks and z_2 = c C[k_2] + c' C[k_wrong] with C[k_2] orthogonal to C[k_wrong]; the argmax is decided by the relative kernel weights at the two positions, which vary with the random permutation pi, so accuracy is strictly below 1. QED.

Non-vacuity: n=4, V=64, C=Sylvester Hadamard, P=2, rho=(0.90,0.999), w=(0.5,0.5), H=1..4 satisfies H1-H7 and attains 1.0 on the diagonal.

Boundary / scope. SCOPE. (i) Distinct keys and strict key/value alternation are essential; with repeated keys the uniqueness of t_h fails and de-duplication is NOT proved here. (ii) 'Exactly H' is under LAST-LAYER readout with no stop mechanism; if intermediate outputs are retained and the H-th is selected, then H_L>=H also suffices (the lower bound H remains, the upper bound disappears). (iii) Orthogonality of C is sufficient but not necessary; only self-inner-product dominance is needed, which is why a Gaussian i.i.d. code also attains 1.0 in the ablation. (iv) Layer normalization q=z/||z|| is a KERNEL-EXTERNAL nonlinearity: rho and theta remain time-invariant, so B4 does not apply. (v) SCOPE CORRECTION (W9). An earlier draft used the absolute-position parity mask r[t]=1[(t-1) even] as H5. That is a HIDDEN FORMAT PRIOR: it is sufficient ONLY when keys always sit at even positions. With arbitrary filler spacing it drops from 1.0 to 0.3438 +/- 0.0314 (5 seeds). H5 is now the explicit 1-bit role tag carried by the input symbol; parity is its special case. The role tag is a fixed deterministic modulation of the INPUT SIGNAL, not of the kernel, so Toeplitz / FFT-IIR duality are preserved. (vi) RELATION TO LINEAR ATTENTION: at rho=1 with a single pole, z_h = q_h^T (sum_t r[t] C[x_(t-1)] C[x_t]^T), which is isomorphic to a linear-attention state; with rho<1 and multiple poles K_h(d)=sum_p w_p rho_p^d is a DECAY-SPECTRUM weighted version. We do NOT claim to leave that family. The contribution is the joint tightness of (order 2, exactly H layers, O(1) state, static Toeplitz kernel, no N x N) -- not a claim of a new paradigm. (vii) This is a SYNTHETIC content-chain retrieval theorem. It does NOT establish language-model parity, perplexity, or generation quality.

Architectural use. Closes the architecture-search corner left open by NT11/NT28: multi-hop content addressing is compatible with FFT/IIR duality after all, at the exact price of one static-LTI layer per hop plus a theta->pi role basis. Predicts three falsifiable observations: single-layer H>=2 must be at chance; H layers must be exactly 1.0 with ZERO training; H+1 layers must FAIL. It also converts the earlier '0.5918 partial construction' into a tight construction and explains the residual as ROLE AMBIGUITY, not capacity.

Witnesses (not the proof). Analytic proof above. Empirical witnesses (NOT the proof): results/chain_analytic_matrix.json (5x5 layer x hop grid), results/chain_analytic_ablation.json, results/chain_analytic_guards.json, results/chain_analytic_learned.json (5 seeds); results/stress_format_prior.json, stress_five_seed.json (W9 correction). Verifier: 17/17 checks pass.

A.32 Lemma R1 - Minimality and irreducibility of the one-bit role tag

Status. SUPPORTING LEMMA, not a promoted core theorem. Hypotheses P1-P4 are stated in full below; the proof is analytic. Verifier: 12/12 checks (witness agreement only).

Premises. P1. A content chain of n pairs (k_p, k_(p+1)) with DISTINCT keys; each pair is presented as two ADJACENT tokens (key immediately followed by value); pair order is arbitrary; between pairs any number of zero FILLER tokens may be inserted. P2. Static LTI kernel K(d)=sum_p w_p rho_p^d > 0, fixed deterministic code C with C C^T = I on the key range, query supplied externally. P3. The layer interface is any FIXED (non-learned, content-independent) linear map followed by normalization; all coefficients of the kernel are time-invariant. P4. Available side information per position t is a binary gate w_t in (0,1); w_t may depend on absolute position t, or on a role tag attached to the input symbol, but NOT on the answer k_H or on the pair permutation in a way that leaks it.

Claim. (a) POSITION-ONLY GATES ARE FORMAT PRIORS, NOT

ARCHITECTURE. Let w_t = 1[(t-1) even] (the v5 parity mask). If each pair is preceded by f_p filler tokens drawn independently with P(f_p odd) >= 1/2, then for every hop h the indicator w at the unique key-successor position is 1 with probability at most 1/2 and is INDEPENDENT of k_h. Whenever it is 0 the true peak is suppressed and z_h carries no information about k_h; hence expected accuracy <= 1/2 + 1/(2 V_K). (b) ONE BIT IS SUFFICIENT AND FORMAT-INDEPENDENT. Let w_t = 1[role(x_(t-1)) = KEY], where each input symbol carries a deterministic KEY / VALUE / FILLER tag. Then for every pair permutation and every filler pattern the NT31 induction gives z_h = c_h C[k_h] with c_h>0, so accuracy is exactly 1 with zero training. (c) CODE-SUBSPACE PARTITION CANNOT REPLACE THE BIT. Suppose we split C into two orthogonal row subspaces C_key, C_val, set A_t = C_key[x_(t-1)], b_t = C_val[x_t], w_t == 1 (no role information), and use any FIXED linear layer interface T. Then accuracy is strictly below 1 on the tight alternating layout. (d) NECESSITY IS LAYOUT-CONDITIONAL. With w_t == 1 and a TIGHT alternating layout, accuracy is below 1. With arbitrary filler spacing, filler tokens implicitly zero the spurious peak and accuracy can exceed 0.95. Therefore 'role information is necessary' must NOT be stated as an unconditional theorem. (e) The role tag acts on the INPUT SIGNAL only; every kernel remains static and Toeplitz, so FFT/IIR duality and O(1) state are preserved.

Proof (a). The p-th key sits at absolute position s_p = sum_(q<p)(f_q+2) + f_p. Its successor is t_p = s_p + 1, and w_(t_p) = 1[(s_p) even]. Because f_p enters s_p additively and P(f_p odd) >= 1/2, the parity of s_p is independent of the pair permutation and of the answer k_h, and P(w_(t_p)=1) <= 1/2. Condition on the event w_(t_p)=0 (probability >=1/2): the h-th matching signal m_h[t_p]=0, so by NT31's base step with distinct keys the only remaining contributions are either zero (filler predecessors) or value-subspace codes orthogonal to q_h; in either case z_h is independent of k_h and the decoded answer is uniform on V_K, giving accuracy 1/V_K. Unconditionally, accuracy <= 1/2*1 + 1/2*(1/V_K) <= 1/2 + 1/(2 V_K). QED.

Proof (b). Replace every occurrence of 'even index' in the NT31 induction by 'role(x_(t-1)) = KEY'. By P1 a KEY tag appears exactly once per pair, and by distinctness k_(h-1) tags exactly one pair, so the match is unique for ANY filler pattern. All other steps are verbatim. QED.

Proof (c). In the tight alternating layout the symbol k_h occurs TWICE: as the VALUE of pair h-1 and as the KEY of pair h. Hence A_t = C_key[k_h] at two distinct positions t_a, t_b. Let T be any fixed linear layer interface and q_(h+1) = T(z_h). Case 1: q_(h+1) has zero inner product with C_key[k_h]; then m_(h+1) == 0 at both positions, z_(h+1)=0, and accuracy is chance. Case 2: the inner product is nonzero; then BOTH positions contribute, and z_(h+1) = K(t_a) C_val[w] + K(t_b) C_val[k_(h+1)], where w is the key of the pair adjacent to pair h-1 in the RANDOM permutation and is independent of k_(h+1). Because C_val[w] and C_val[k_(h+1)] are orthogonal, argmax selects k_(h+1) iff K(t_b) > K(t_a), an event determined by positions, not by the answer; over random permutations this fails with strictly positive probability. Hence accuracy < 1 for every fixed T. QED.

Proof (d). Tight layout gives two nonzero peaks => (c) applies. In a gapped layout the token following a VALUE is a zero FILLER, whose b_t is the zero vector, so the spurious peak contributes nothing and only the true peak survives; accuracy can approach 1. Thus the necessity of the bit is a property of the LAYOUT, not a universal law. QED.

Proof (e). w_t multiplies A_t = C[x_(t-1)], i.e. the input signal u_t. The impulse response K(d) is unchanged and depends only on the delay; Toeplitz holds, so a single FFT realizes the same operator as the IIR scan. The state is P vectors per layer, independent of L. QED.

NON-VACUITY: n=4, V=64, VK=32, C = Sylvester Hadamard, f_p ~ Uniform(0..4), H=2, H_L=2 satisfies P1-P4; (a) witnesses 0.3438 +/- 0.0314 (5 seeds), (b) witnesses 1.0000 on all four layouts, (c) witnesses 0.8691, (d) witnesses 0.8621 (none, tight) vs 0.9609 (none, gapped).

Boundary / scope. SCOPE. (i) This is a SUPPORTING LEMMA, not a 32nd headline theorem; the v5 main paper promotes exactly six core theorems. (ii) The upper bound in (a) is an expectation over the filler distribution, not a worst-case bound for every fixed filler pattern; a pattern with all keys at odd positions would give exactly chance, while one with all keys even gives 1. (iii) The 1-bit tag is minimal among SIDE-CHANNELS but we do NOT prove that no architecture outside P1-P4 (e.g. learned input-dependent gates, extra recurrence, or retaining intermediate outputs) could work without it. (iv) 'KEY / VALUE / FILLER' is a SYMBOL-TYPE annotation, not an answer leak: it does not reveal which key is queried or what k_H is. (v) Filler tokens are zero one-hots; their presence is itself structural information, which is precisely why (d) refuses an unconditional necessity claim. (vi) Synthetic content-chain task only; no LM claim. (vii) WITNESS CAVEAT: the first implementation of the (c) candidate used (A_prev @ Ck.T @ Ck), which is the projection Ck^T Ck rather than the key code. That bug was found and fixed; the recorded 0.8691 is from the corrected implementation and equals the no-role baseline, confirming the subspace partition adds nothing.

Architectural use. Converts the most damaging potential rejection ('the construction hides a task-format prior') into an auditable scope correction. Justifies withdrawal W9 and the H5 revision of NT31. Gives a concrete engineering rule: emit a role tag with each symbol (or use separate key/value symbol ids) instead of relying on absolute position parity. Also explains why the earlier 0.5918 two-layer result and the 0.8691 no-role baseline share the same root cause: role ambiguity, not capacity.

Witnesses (not the proof). Analytic proof above. Empirical witnesses (NOT the proof): results/stress_format_prior.json, stress_five_seed.json, stress_duplicate.json, stress_precision.json, stress_throughput.json, stress_adjudication.json. Code: tools/native_probe/chain_stress_probe.py.

A.1-A.30 Retained full ledger (NT01-NT30). Each entry below states its premises, universal claim, complete proof, boundary and architectural use. These are the supporting evidence chain behind T1/T3-T6 and P1-P8.

A.1 NT01 - Tight minimum tensor order for exact single-hop retrieval

Premises. Static LTI kernel K_n(d)>0 for every delay d>=0 (strictly positive); history contains n>=2 pairs (k_p,v_p) with independently assignable one-hot values; q=k_(j*); code rows satisfy <Code[a],Code[b]>=R*delta_(a,b); read-out is z_n = sum_(p,r) w_(p,n) q_r s_(p,r,n,L) with s_(p,r,n,t)=rho_(p,n)s_(p,r,n,t-1)+a_t[r]b_t[n].

Claim. (a) A vector-valued (order-1) state that is a fixed-dimensional linear summary of the history cannot implement the retrieval, for ANY read-out. (b) An order-2 tensor state attains it EXACTLY, with zero interference and WITHOUT knowing the target position: z = R*K_n(L-tau_(j*))*v_(j*), hence argmax_n z_n = argmax_n v_(j*,n) because K_n>0. Therefore the minimum tensor order is exactly 2, and the bound is TIGHT.

(a) Let A=L(v_1..v_n) in R^d be the summary, L: R^(nd)->R^d linear and independent of q. rank(L)<=d<nd for n>=2, so dim ker L>=nd-d>0. Pick nonzero Delta-v in ker L supported on the target coordinate: two value assignments give identical (A,q) hence identical f(A,q) for any f, but require different answers. Contradiction.

(b) Unroll the recurrence: s_(p,r,n,L)=sum_tau rho_(p,n)^(L-1-tau) a_tau[r] b_tau[n]. Contract with q_r and w_(p,n): z_n = sum_tau (q . k_tau) K_n(L-1-tau) v_(tau,n) with K_n(d)=sum_p w_(p,n) rho_(p,n)^d. By code orthogonality q . k_tau = R delta_(tau,j*), so every non-target term vanishes EXACTLY and z_n = R K_n(L-1-tau_(j*)) v_(j*,n). Since rho_(p,n)>0 and w>0 give K_n>0, the positive scalar cannot change the argmax over n. No position information is needed at read-out. Hence order 2 suffices.

Tightness: (a) forbids order 1 and (b) achieves order 2. QED

Boundary. Requires K_n(d)>0; a damped-oscillator kernel sin((tau+1)theta) changes sign and is NOT admissible. Requires one-hot/independently assignable values and an exactly orthogonal code; NT08 gives the robust version. Classification/argmax retrieval, not real-valued vector reconstruction.

Architectural use. Explains why the general real-pole dictionary beats the oscillator grid, and turns 'TSC works' into a tight optimality statement rather than an empirical accuracy. *Source:* Section 4 TSC; core/arch/sgc.py general second-order IIR dictionary

A.2 NT02 - Strict-positivity separation of the two pole families

Premises. Second-order IIR y_t = a y_(t-1) + b y_(t-2) + c u_t with characteristic roots either (i) two positive reals 0<r2<=r1<1, a=r1+r2,

b=-r1*r2, or (ii) a conjugate pair rho*e^(+-i theta), 0<rho<1, 0<theta<pi, a=2 rho cos(theta), b=-rho^2.

Claim. Family (i) has h(tau)>0 for every tau>=0; family (ii) has h(tau)<=0 for some tau<=ceil(pi/theta). Hence the two impulse-response families are DISJOINT, and membership is decided exactly by inspecting the sign sequence of h(0..L-1) in O(L) (zero misclassification in exact arithmetic). Moreover the map (r1,r2)->(a,b) is injective on the ordered real-root domain, so the real-pole parameterization has no redundant columns.

(i) h(tau)=sum_(j=0)^(tau) r1^(tau-j) r2^(j) = (r1^(tau+1)-r2^(tau+1))/(r1-r2) for r1!=r2, and (tau+1)r^(tau) for r1=r2=r. Every term is strictly positive, so h(tau)>0.

(ii) h(tau)=rho^tau sin((tau+1)theta)/sin(theta) up to a positive normalization. Let tau0=ceil(pi/theta). Then (tau0+1)theta >= pi+theta > pi and (tau0+1)theta <= pi+2theta < 3pi. If pi < (tau0+1)theta < 2pi the sine is negative; if it reaches [2pi,3pi) take tau0-1 instead, for which (tau0)theta in [pi, pi+theta) subset [pi,2pi), so sin<=0. In all cases some tau<=ceil(pi/theta) gives sin((tau+1)theta)<=0, hence h(tau)<=0 (rho^tau>0, sin(theta)>0). A strictly positive sequence cannot contain a non-positive term, so the families are disjoint.

Injectivity: (a,b) determines the monic polynomial z^2 - a z - b, whose roots are exactly (r1,r2); ordering r2<=r1 makes the pair unique. Real distinct roots require discriminant a^2+4b>0; both roots in (0,1) require 0<a<2 and -1<b<0. QED

Boundary. Root r1=r2 is a double real root (still strictly positive), handled by the limit formula. theta=pi is excluded by the architecture's theta clamp. Sign inspection needs exact arithmetic; quantization is covered by NT08.

Architectural use. Gives a provable, O(L)-checkable discriminant between the oscillator basis and the general second-order dictionary, and shows the latter has no parameter redundancy (unlike theta->0 collinearity, NT06). *Source:* core/arch/sgc.py _init_oscillator_grid / general second-order dictionary

A.3 NT03 - Provable separation of rho<1 multi-pole TSC from the linear-attention family

Premises. TSC read-out kernel K_n(d)=sum_(p=1)^(P) w_(p,n) rho_(p,n)^(d) with P>=1, 0<rho_(p,n)<1, w_(p,n)>0, all rho_(p,n) distinct. Linear-attention (single-pole, rho=1) kernel is the constant K(d)=1.

Claim. The span of (K_n : n) contains NO nonzero constant function, and the constant function is not in the closure of that span in l^infty(N). Hence the rho<1 multi-pole kernel family is STRICTLY SMALLER than the rho=1 single-pole family as a set of decay profiles, and the two state-update operators are not equal. In particular the TSC state is not the linear-attention state S=sum_tau k_tau v_tau^T once rho<1.

Any nonzero combination F(d)=sum_n c_n K_n(d) is a finite sum of exponentials with bases in (0,1). Let rho_max<1 be the largest base; then F(d)=Theta(rho_max^d) -> 0 as d->infinity. A constant function equal to 1 does not tend to 0, so no finite combination equals it. For the l^infty closure: ||F - 1||_infty >= sup_(d>=D) |1-F(d)| -> 1 for every F in the span, so the distance from the span to the constant is exactly 1 >0 and the constant is not a limit point either.

Consequently the rho=1 update s_t=s_(t-1)+u_t (constant kernel, state S=sum k_tau v_tau^T) and the rho<1 update s_t=rho s_(t-1)+u_t (decaying kernel) generate different linear operators on the same tensor signal; they cannot be identified by any change of basis that preserves the impulse response. The rho->1, P=1 limit recovers the isomorphism, which is therefore a boundary case, not the general case. QED

Boundary. This is a separation of KERNEL FAMILIES / state-update operators, not a claim that the full models have different Turing-complete expressivity, nor a claim of empirical superiority. It does not assert that TSC is outside every linear-attention VARIANT (e.g. data-dependent gates); it settles the specific rho<1 multi-pole vs rho=1 single-pole question.

Architectural use. Directly answers the reviewer's point 3: instead of leaving the TSC / linear-attention relation 'to the reader', we give a provable non-equivalence while still recording the rho->1 isomorphism. *Source:* Section 4 remark (rho->1 isomorphism) + core/theory/smm_sgc.py

A.4 NT04 - Deterministic zero-crosstalk codes for arbitrary V are 2-optimal

Premises. V>=1 arbitrary integer; code rows in R^R with zero crosstalk.

Claim. (a) Zero crosstalk among V nonzero rows forces R>=V. (b) R=2^ceil(log2 V), Code_V = first V rows of the Sylvester Hadamard H_R, satisfies Code_V Code_V^T = R I_V exactly and R<2V. Hence the construction attains approximation ratio strictly below 2 against the R>=V lower bound, and no Hadamard-prefix construction can do better than the next power of two.

(a) V mutually orthogonal nonzero vectors in R^R are linearly independent, so V<=R.

(b) Induction on the Sylvester recursion H_(2m)=[[H_m,H_m],[H_m,-H_m]] gives H_R H_R^T = R I_R; the leading V-by-V principal submatrix of that identity is exactly Code_V Code_V^T = R I_V. Since R=2^ceil(log2 V) < 2^(log2 V + 1) = 2V for V>1 (and R=V=1 for V=1), R/V < 2. Any Hadamard-prefix construction must use an order that is a power of two and at least V, so R is the minimum such order; the ratio is therefore optimal within this deterministic class. QED

Boundary. Existence + explicit construction + optimality within Hadamard prefixes. Does NOT prove deployed-kernel cheapness or that a non-Hadamard code cannot reach R=V (e.g. an orthonormal V-by-V basis would, but is not a deterministic +/-1 code).

Architectural use. Closes the V>>64 construction question with a clean 2-approximate bound. *Source:* CT5-L2 / core/theory/smm_sgc.py DETERMINISTIC_CODE

A.5 NT05 - State-vocabulary lower bound and end-to-end tightness

Premises. Static LTI tensor state s_(p,r,n) with P pole channels, R code rows, N value channels; exact zero-crosstalk retrieval of V token classes under NT01's premises.

Claim. Any such construction requires at least P*V*N scalar state registers, i.e. Omega(V*N). TSC with the NT04 code uses P*R*N < 2 P*V*N registers, so it attains the lower bound within a factor 2 [illegible] code dimension, state size and retrieval exactness are simultaneously tight.

Zero crosstalk requires R>=V mutually orthogonal code rows (NT04a). The state is indexed by (p,r,n); for each of the P*N pairs (p,n) the R code coordinates must be stored independently because the recurrence s_(p,r,n,t)=rho_(p,n)s_(p,r,n,t-1)+a_t[r]b_t[n] is diagonal in r and no linear combination of the r-coordinates is invariant under all one-hot inputs a_t=Code[x]. Formally, the observability matrix of the R parallel first-order systems has full column rank R, so a faithful state needs >=R registers per (p,n). Hence total >=P*R*N>=P*V*N. With R<2V from NT04b the construction uses <2P*V*N: ratio <2. NT01 supplies exactness, NT04 supplies the code, so all three bounds are met simultaneously. QED

Boundary. Lower bound is for faithful (injective-in-r) static LTI tensor states with a fixed code. Compressed or learned non-tensor states are outside the premise. The factor 2 is a guarantee, not optimality of the constant.

Architectural use. Turns the TSC construction into a tight state-complexity result instead of a single accuracy number on V=64. *Source:* Section 4 cost remark (56,320 B state) + NT01/NT04

A.6 NT06 - Quantitative rank-deficiency rate of near-zero frequency grids

Premises. Dictionary columns f_m(tau)=rho^tau sin((tau+1)theta_m), 0<rho<1, theta_1<theta_2<...<theta_M, Delta=theta_2-theta_1, Gram matrix G_(ij)=<f_i,f_j> over tau>=0.

Claim. cond(G) >= 2*sin^2(theta_1) / (Delta^2 * A(rho)), where A(rho)=sum_(tau>=0) rho^(2tau)(tau+1)^2 = (1+rho^2)/(1-rho^2)^3 < infinity. Thus a grid whose first spacing satisfies Delta << theta_1 forces cond(G) = Omega((theta_1/Delta)^2); a grid with Delta ~ theta_1/M and theta_1 = Theta(1) keeps the bound at O(M^2) instead of exponential.

Use the Rayleigh quotient with the vector e_1-e_2: sigma_min(G) <= (1/2)||f_1-f_2||^2. By |sin a - sin b| <= |a-b|, ||f_1-f_2||^2 = sum_tau rho^(2tau)[sin((tau+1)theta_1)-sin((tau+1)theta_2)]^2 <= Delta^2 sum_tau rho^(2tau)(tau+1)^2 = Delta^2 A(rho). The closed form follows from sum n^2 x^n = x(1+x)/(1-x)^3 with x=rho^2 and index shift. Also sigma_max(G) >= G_(11) = ||f_1||^2 >= rho^0 sin^2(theta_1) =

sin^2(theta_1). Hence cond = sigma_max/sigma_min >= 2 sin^2(theta_1)/(Delta^2 A(rho)). For logfreq grids with theta_1 extremely small and Delta proportional to theta_1*(r-1), the two small quantities nearly cancel in the RATIO but A(rho) explodes as rho->1; for linear grids theta_1=Theta(1) and Delta=Theta(1/M), giving cond >= Theta(1/M^2 * 1/A(rho)) while the measured value is O(1), i.e. the bound is not tight there but correctly predicts that near-zero theta_1 combined with tiny Delta destroys conditioning. QED

Boundary. A LOWER bound on cond (a badness certificate), not an exact value and not an upper bound. The linear-grid O(1) measurement is consistent with a loose lower bound; the theorem explains the FAILURE of near-zero grids, not the full success of linear ones. Condition numbers remain observations.

Architectural use. Upgrades the earlier 'direction only' asymptotic into a checkable rate, and explains why theta_min=pi/8192 collapses capacity while a linear grid restores it. *Source:* Prop8 / docs/frequency_grid_fix.md / core/arch/sgc.py init_mode

A.7 NT07 - Precision-length-decay trilemma with an interior optimal rho

Premises. First-order streaming state s_t=rho s_(t-1)+u_t, 0<rho<1, |u_t|<=U; each update is quantized with absolute step error <= delta, delta/U < 1; a distant contribution at delay L has amplitude rho^(L-1) U and must exceed c times the accumulated error floor (c>1 a fixed separation constant); c*delta/U < 1.

Claim. (a) The accumulated error is bounded by e_infty <= delta/(1-rho). (b) The resolvable length obeys L(rho) <= 1 + ln((1-rho)U/(c*delta)) / |ln rho|. (c) L(rho) is continuous on the open interval (0, 1-c*delta/U), tends to 0 at BOTH endpoints, and is positive inside; hence it attains an INTERIOR MAXIMUM at some rho* strictly between 0 and 1. Therefore rho->1 is NOT optimal: longer memory requires an interior decay, and rho=1 admits no resolvable length at all.

(a) The error obeys e_t <= rho e_(t-1) + delta; by induction e_t <= delta sum_(k=0)^(t-1) rho^k < delta/(1-rho).

(b) Resolvability requires rho^(L-1) U > c*delta/(1-rho), i.e. (L-1) ln rho > ln(c*delta/((1-rho)U)). Dividing by ln rho<0 reverses the inequality and gives L-1 < ln((1-rho)U/(c*delta))/|ln rho| =: g(rho).

(c) g requires (1-rho)U > c*delta, i.e. rho < 1 - c*delta/U. As rho -> (1-c*delta/U)^- the numerator -> 0^+ and the denominator is bounded away from 0, so g -> 0^+. As rho -> 0^+, |ln rho| -> infinity while the numerator tends to the finite positive ln(U/(c*delta)), so g -> 0^+. g is continuous and strictly positive somewhere inside (e.g. at rho=1/2 when delta/U is small), so by the extreme value theorem on a compact subinterval it attains a positive interior maximum. At rho=1 the error floor delta/(1-rho) is unbounded, so no L satisfies (b). QED

Boundary. CONDITIONAL on a uniform absolute quantization-error model (same premise as the perturbation bound). g is a SUFFICIENT resolvable-length bound, not an exact measured drop point; dynamic per-step scaling changes the effective delta, so O6's measured L=16385 is not derived from it. The existence of rho* is proved, its numerical value is an observation.

Architectural use. The flagship quantitative result: proves that the best decay is interior, explains rho=0.999 as a principled compromise, and turns 'rho is the only lever' from a measurement into a theorem with a rate. *Source:* O6 physics probes + Appendix A.7 perturbation bound + T12/T13

A.8 NT08 - Robust exact retrieval under bounded code perturbation

Premises. NT01's architecture, but the code satisfies Code Code^T = R I_V + E with ||E||_infinity <= epsilon; all kernels K_n(d) in [K_min,K_max] with K_min>0; values are one-hot with |v|<=1; at most L-1 non-target terms.

Claim. The retrieval remains exactly correct (argmax unchanged) whenever epsilon < R*K_min / ((L-1)*K_max). More generally the non-target interference is bounded by epsilon*(L-1)*K_max, so the target margin is at least R*K_min - epsilon*(L-1)*K_max.

Write q.k_tau = R delta_(tau,j*) + E_(q,tau). Then z_n = R K_n(L-1-tau_(j*)) v_(j*,n) + sum_(tau != j*) E_(q,tau) K_n(L-1-tau) v_(tau,n). The target coordinate of the one-hot value has amplitude >= R K_min. Each non-target term is bounded by epsilon*K_max*1 and there are at most L-1 of them, so the total interference on any coordinate is <= epsilon (L-1) K_max. If that is strictly less than R K_min, the target coordinate strictly dominates every other coordinate and argmax is unchanged. The stated inequality is exactly this condition. QED

Boundary. A SUFFICIENT condition, not necessary; assumes bounded kernels and one-hot values. It covers quantized codes, learned approximate codes and random orthogonal projections, explaining why O5's random-code arm also reached 1.0000 without any retraining.

Architectural use. Makes NT01 deployable: the deterministic Hadamard code is not a fragile requirement, only a bounded-crosstalk one. *Source:* Section 4 code remark + O5 random orthogonal code control

A.9 NT09 - Rounding-order crossover between the FFT and IIR paths

Premises. Standard floating-point model with unit roundoff eps; an L-point radix-2 FFT has log2(L) butterfly stages with relative error at most c1*eps per stage; the stable second-order IIR has steady-state relative error at most c2*eps/(1-rho), independent of L; c1,c2>0 fixed.

Claim. The IIR error bound is L-independent while the FFT bound grows as c1*eps*log2(L). They cross at L* = 2^(c2/(c1(1-rho))). For every L < L* the streaming path has the smaller worst-case bound; for rho close to 1, L* is astronomically large, so on all realizable sequence lengths the IIR bound is never worse.

FFT: after log2 L stages, (1+c1 eps)^(log2 L) - 1 <= c1 eps log2 L + O(eps^2). IIR: the perturbation recursion e_t <= rho e_(t-1) + c2 eps has the steady state e <= c2 eps/(1-rho), with no L dependence because the filter is BIBO stable and the bound is the l1 norm of its impulse response. Equating c1 eps log2 L = c2 eps/(1-rho) cancels eps and gives log2 L* = c2/(c1(1-rho)), i.e. L* = 2^(c2/(c1(1-rho))). Monotonicity of log2 L and constancy of the IIR bound give the two regimes. QED

Boundary. CONDITIONAL on the stated floating-point model and on WORST-CASE BOUNDS, not measured error. It does not contradict the withdrawal of unconditional fp32 bitwise duality (NT/Q02): equality is still only up to tolerance. It is a selection criterion between execution orders, not a speed claim.

Architectural use. Supplies the missing numerical-selection rule for the dual-form theorem: which path to trust at a given L and rho. *Source:* Thm1 remark + backend 16-op contract + Appendix A.7

A.10 NT10 - Per-channel time-invariance preserves the duality

Premises. TSC recurrence s_(p,r,n,t)=rho_(p,n) s_(p,r,n,t-1)+u_t[r,n] with rho_(p,n) depending on the pole channel p and value channel n but NOT on time t or code index r; coefficient tensor w_(p,n) likewise static.

Claim. Each (p,n) pair is an independent LTI system with transfer function H_(p,n)(z)=1/(1-rho_(p,n) z^(-1)). The full operator is BLOCK Toeplitz (Toeplitz in t for each channel, diagonal in (p,n)), so a single batched FFT diagonalizes it channel-wise and the streaming recurrence realizes the same operator. The state has P*R*N scalars, independent of L. Consequently a per-channel coefficient tensor of shape (M,D) is architecturally admissible without breaking any purity constraint.

For fixed (p,r,n), unrolling gives s_(p,r,n,t)=sum_tau rho_(p,n)^(tau) u_(t-tau)[r,n], whose impulse response depends only on the delay tau. Stacking the P*N independent channels gives a block-diagonal-in-(p,n) operator whose t-blocks are Toeplitz; by NT12 (applied per channel) the parameter must be constant in t, which it is. A Toeplitz operator is diagonalized by one DFT per channel, so one batched FFT suffices; the recurrence is the same rational function computed in the time domain. The state count is P*R*N with no L term. QED

Boundary. Requires rho and w independent of t. It does NOT permit input-dependent rho_t (that is the SFSC/LTV case, excluded by NT12/Q09). The constant P*R*N may be large (56,320 B observed); O(1) is asymptotic.

Architectural use. Upgrades the v4 finding (coef was artificially flattened from (32,2048) to 32 scalars) from a script simplification into a theorem about what the architecture permits. *Source:* core/arch/sgc.py coef (M,D) + T7 / Section 4

A.11 NT11 - Layer-count lower bound for multi-hop retrieval

Premises. A stack of L layers, each a polynomial map of total degree at most d in its inputs (a bilinear contraction has d=2); skip connections do not raise degree; the target H-hop output contains H key-match factors and one value factor, i.e. total degree >= H+1 in the history variables.

Claim. Any such stack needs L >= ceil(log_d(H+1)) layers. For bilinear layers L >= ceil(log2(H+1)); in particular H=2 already requires L>=2, so a single contraction layer cannot implement two-hop retrieval. This is a NECESSARY condition; no sufficient construction is asserted.

The composition of polynomials of degrees d_1..d_L has degree at most the product d_1...d_L <= d^L (degree cannot increase under addition or skip connections). The H-hop target is a multilinear form of degree H+1 in (k,v): H factors of the form <phi(q),psi(k)> plus one value factor. A polynomial of degree < H+1 cannot equal a nonzero degree-(H+1) form on an open set. Hence d^L >= H+1, i.e. L >= log_d(H+1); taking ceil gives the integer bound. For d=2, H=2: log2(3)=1.58 -> L>=2. QED

Boundary. NECESSARY only. The degree bound is an upper bound on what L layers CAN reach; reaching it needs a construction (untested). Per-position nonlinearities of degree >2 change d. The measured H4_multihop=0.0322 is a finite-sample observation, not a theorem conclusion.

Architectural use. Gives the stacking roadmap a concrete quantitative target instead of 'try more layers', and explains the two-hop collapse structurally. *Source:* T15 / L7 / O5 H4_multihop; core/arch stacking

A.12 NT12 - Toeplitz iff time-invariant coefficients, for arbitrary order and channels

Premises. y_t = sum_(j=1..P) a_(j,t) y_(t-j) + c_t u_t with c_t != 0 for all t; P>=1 arbitrary; optionally with an independent copy for each of C channels (a_(j,t)^((c)), c_t^((c))).

Claim. The impulse response h(t,s) depends only on t-s IFF every coefficient a_(j,t) and c_t is constant in t. With C independent channels the full operator is block Toeplitz IFF this holds for every channel. Necessity fails at the first delay where c_t=0, so the theta-clamp / nonzero-c premise is essential, not decorative.

Necessity by induction on the delay k. k=0: h(s,s)=c_s must be s-independent, so c_t=c. Assume a_1..a_(k-1) are constant. At delay k, h(s+k,s) = a_(k,s+k) c + Phi(a_1..a_(k-1),c), where Phi is a polynomial with s-independent coefficients. Toeplitz forces h(s+k,s) independent of s, and c != 0 gives a_(k,t) constant. Induct to k=P. Sufficiency: constant coefficients give the LTI transfer function H(z)=c/(1-sum_j a_j z^(-j)), so h(t,s)=h(t-s). For C channels apply the same argument to each diagonal block; off-diagonal blocks are zero. If c_t=0 for some t, the k=1 step cannot isolate a_(1,t), and a counterexample with time-varying a_1 but Toeplitz h exists (the input never enters at that step). QED

Boundary. The equivalence concerns THIS convolution/FFT path; it does not assert nonexistence of other sub-quadratic algorithms. c_t != 0 is guaranteed in SGC by sin(theta)>0 under the theta clamp; removing the clamp invalidates the necessity direction.

Architectural use. The precise equivalence-strengthening the reviewer asked to foreground; now covers arbitrary order and per-channel grids, and identifies exactly where the premise can break. *Source:* Thm4 + T11 + Q09/Q11; core/arch/sgc.py theta_clamp_floor

A.13 NT13 - Arbitrary-read-out impossibility with a sharp quantitative boundary

Premises. EXACTLY four premises: (i) the history enters the final computation only through a fixed-dimensional LINEAR summary A=L(v_1..v_n) in R^d that is independent of q; (ii) q enters only at the current position; (iii) values are independently assignable; (iv) at least two distinguishable targets exist. Read-out f(A,q) is arbitrary (linear or nonlinear). Key linear independence is NOT required.

Claim. (a) Content addressing is impossible for any f. (b) Quantitatively dim ker L >= nd - d. (c) For n=1 the kernel can be zero, so premise (iv) is NECESSARY and cannot be dropped. (d) TSC escapes by changing the tensor order of the signal and the read-out, NOT by contradicting (a).

(a,b) rank(L)<=d, so dim ker L = nd - rank(L) >= nd-d > 0 for n>=2. Choose nonzero Delta-v in ker L supported on a coordinate of the target value (possible by (iii),(iv)). Two assignments v and v+Delta-v give the same A and the same q, hence the same f(A,q), but require different answers. Contradiction for every f. Linear read-out follows by subtracting two queries. (c) n=1 gives nd=d; a random square L is full rank almost surely, so ker=0 and (a) no longer follows -- the premise is sharp. (d) NT01 uses an order-2 tensor state, which is not a fixed vector summary, so (i) is false; the theorem is inapplicable, not violated. QED

Boundary. DOES NOT extend to deep stacks with per-position nonlinearities that mix A and q, to input-dependent kernels (NT12 instead), or to outer-product states (NT01). Corollary B1.1 ('requires a pairwise term') is an EXPLANATORY FRAMEWORK, NOT A THEOREM, and is not a proved sufficient or necessary characterization of all escape mechanisms.

Architectural use. The strongest boundary theorem, with an explicit four-premise contract that must appear in the abstract and conclusion to prevent misreading. *Source:* Thm5 + T10 + CT4; Appendix A.5

A.14 NT14 - Information-Gradient Duality (with the softening trap)

Premises. An architecture has (a) an online state s_L=F(x_(1:L)) in R^D with D independent of L and F continuous; (b) an episodic bank of C EXACT copies read by hard top-k selection S_k(q) followed by continuous softmax weights; (c) no N x N score matrix, no resident KV cache; (d) the task requires exact retrieval of an arbitrary one of L history items, |X|>1, L unbounded.

Claim. No single channel is simultaneously (I) differentiable, (II) covering all L items, and (III) losslessly retrieving any one. The three horns are exclusive AND the natural repair is self-defeating: replacing hard top-k by a differentiable soft aggregate over all C copies re-triggers NT13's kernel impossibility.

(i) Differentiable + covering forces lossy: F must be injective on X^L, but a B-byte state has at most 2^(8B) configurations and |X|^L > 2^(8B) for L > 8B/log2|X| (pigeonhole).

(ii) Hard top-k: on the tie-free open set the ordering is locally constant, so d 1[j in S_k]/d theta = 0 a.e. and the Jacobian support is contained in S_k of size k; the C-k unselected items contribute exactly zero to any gradient update. Exact on k slots, fails (I) for selection and (II) because k<L.

(iii) SOFTENING: m_soft(q)=sum_(j=1..C) alpha_j(q) v_j is differentiable and covers all C, but for fixed q the map (v_1..v_C) in R^(Cd) -> m_soft in R^d is LINEAR with rank <= d, so dim ker >= Cd-d > 0 for C>=2. A nonzero Delta-V in that kernel leaves m_soft and q unchanged while changing the target value, so any f(m_soft,q) gives identical output for distinct required answers. This is exactly NT13. Hence (I)+(II) again forces NOT (III).

The three repairs are mutually exclusive. QED

Boundary. Does not forbid hard top-k retrieval of k items (the L2 pointer sidecar), nor a zero-order path (NT15). It says nothing about empirical LM quality. Numerical checks are finite identities, not the proof.

Architectural use. The hardest structural theorem: turns 'gradient vanished' into a closed-loop impossibility with a self-defeating repair. *Source:* CT6 / core/theory/smm_sgc.py; memory.py topk, smm.py @torch.no_grad

A.15 NT15 - Zero-order self-evolution: gate validity and information rate

Premises. NT14 holds; online answer path has no autograd graph; consolidation writes to weights only through a no-regression gate comparing paired-bootstrap one-sided CI lower bounds l_alpha (level 1-alpha) against -|tau| over m metrics, rejecting if ANY fails; an independent held-out verifier is supplied (otherwise writeback is refused); at most w insights of b bits are committed per round; each read selects k items.

Claim. (a) P(false commit | true regression) <= alpha for one metric and <= m*alpha for m metrics (<= alpha under Bonferroni alpha/m). (b) I(history ; weight increments over T rounds) <= T*w*b bits, INDEPENDENT of L and C; instantaneous bandwidth <= k*b bits/round. (c) The path MUST be zero-order. (d) A non-decreasing ACCEPTED score does NOT imply monotone real capability.

(c) NT14(ii) zero Jacobian for unselected items, NT14(iii) softening destroys exactness, and the no_grad online path has no graph: no gradient

can carry long-context experience into the weights, so the only admissible channel is a zero-order accept/reject test on verifier outputs.

(a) Coverage: P(l_alpha <= Delta_true) >= 1-alpha, hence P(l_alpha > Delta_true) <= alpha. A false commit requires l_alpha >= -|tau| > Delta_true, i.e. l_alpha > Delta_true, exactly the coverage-failure event. Union bound over m metrics gives m*alpha.

(b) By data processing, I(H;Delta theta_t) <= H(insights_t, verifier_t) <= O(w*b). Summing over T rounds gives T*O(w*b). L and C do not appear because NT14(ii) bounds each read to k selected items; the instantaneous bound is k*b.

(d) The gate proves NO-REGRESSION, a necessary condition for writeback; improvement requires a separate held-out evaluation. The upstream docstring states exactly this. QED

Boundary. (a) CONDITIONAL on paired i.i.d. samples, bootstrap consistency and a non-default verifier; with the default constant verifier the loop refuses writeback, so the gate is inactive, not contradicted. (b) is an UPPER BOUND, not a claim the loop reaches it. Monte-Carlo checks are ILLUSTRATION ONLY. T16's scalar-monotonicity claim is WITHDRAWN (W7).

Architectural use. First rate bound for the SMM x SGC self-evolution path; shows long context buys inference-time retrieval, NOT a faster learning rate. *Source:* CT7 / dream/evaluation.py regression_gate, loop.py VERIFY, hooks.py

A.16 NT16 - Exact Turan identity and log-convexity of the real-pole dictionary

Premises. General second-order IIR y_t = a y_(t-1) + b y_(t-2) + u_t with characteristic roots 0 < r2 <= r1 < 1, so a = r1+r2, b = -r1*r2; impulse response h(tau) with h(0)=1, h(-1)=0.

Claim. (a) EXACT IDENTITY: h(tau)^2 - h(tau-1) h(tau+1) = (r1*r2)^tau > 0 for every tau >= 1, including the double-root case r1=r2=r where it becomes r^(2tau). Hence h is STRICTLY log-convex. (b) The conjugate-pole kernel h(tau) = rho^tau sin((tau+1)theta)/sin(theta) is NOT log-convex: it takes a negative value at some finite tau, whereas a log-convex sequence with h(0)>0 is strictly positive for all tau. (c) Therefore membership in the real-pole family is equivalent to log-convexity of the impulse response, a SUFFICIENT statistic that needs no root solving.

(a) Binet form: h(tau) = (r1^(tau+1) - r2^(tau+1))/(r1-r2) for r1>r2, and h(tau) = (tau+1) r^tau for r1=r2=r. For r1>r2 set A=r1^(tau), B=r2^(tau), p=r1, q=r2 and expand h(tau)^2 - h(tau-1)h(tau+1). Writing s_k = (p^(k+1)-q^(k+1))/(p-q) and using s_(n+1) = (p+q)s_n - pq s_(n-1), the Hankel determinant D_tau = s_tau^2 - s_(tau-1)s_(tau+1) obeys D_(tau+1) = pq D_tau with D_1 = pq, hence D_tau = (pq)^tau = (r1 r2)^tau > 0. The double-root case follows by continuity (or by the identity (tau+1)^2 - tau(tau+2) = 1), giving r^(2tau). Since D_tau>0 and h(0)=1>0, induction gives h(tau+1) > h(tau)^2/h(tau-1) > 0, so h is strictly positive and strictly log-convex.

(b) By NT02 there exists tau0 <= ceil(pi/theta) with sin((tau0+1)theta) <= 0, hence h(tau0) <= 0. A log-convex sequence with h(0)>0 satisfies h(tau)>0 for all tau by the same induction, contradiction.

(c) The two families are exhaustive for stable second-order roots (discriminant sign), and (a)/(b) separate them by a checkable inequality on the impulse response alone. QED

Boundary. Normalization h(0)=1 (c=1) is assumed; a general c>0 scales all h by c and leaves D_tau = c^2 (r1r2)^tau > 0, so log-convexity is invariant. theta=pi excluded by the architecture's clamp. The Turan identity is exact over R; finite precision is NT08/NT25.

Architectural use. Gives the STRUCTURAL reason the general second-order dictionary beats the damped-oscillator grid on content addressing (NT01 needs K_n>0), and turns NT02's O(L) sign scan into a log-convexity certificate. It explains why the 'oscillatory basis' was the wrong inductive bias. *Source:* core/arch/sgc.py general second-order IIR dictionary + NT02

A.17 NT17 - O(1) three-sample identifiability and pole-family discriminant

Premises. A stable second-order recursion with impulse response normalized to h(0)=1; the first three samples h(0),h(1),h(2) are observed exactly.

Claim. (a) The recursion coefficients are recovered EXACTLY and uniquely: a = h(1)/h(0), b = (h(2) - a h(1))/h(0). (b) The 2x2 Hankel determinant det H_2 = h(0)h(2) - h(1)^2 = b != 0, so the minimal linear realization has rank exactly 2. (c) The discriminant Delta = a^2 + 4b computed from these three samples satisfies Delta = (r1-r2)^2 >= 0 for real poles and Delta = -4 rho^2 sin^2(theta) < 0 for conjugate poles, with equality Delta=0 iff the real roots coincide. Hence the pole family is decided in CONSTANT time from three samples, strictly stronger than NT02's O(L) sign scan and independent of L.

(a) From y_t = a y_(t-1) + b y_(t-2) + u_t with the unit impulse, h(0)=1, h(1)=a h(0)=a, h(2)=a h(1)+b h(0)=a^2+b. Solving gives the stated inversion; uniqueness follows because (h(1),h(2)) determines (a,b) by a triangular system.

(b) det H_2 = h(0)(a^2+b) - a^2 = b. Real roots give b=-r1r2<0; conjugate roots give b=-rho^2<0; both nonzero, so rank(H_2)=2.

(c) The characteristic polynomial is z^2 - a z - b with roots (a +- sqrt(a^2+4b))/2. For real roots r1,r2 we have a=r1+r2 and b=-r1r2, so a^2+4b=(r1+r2)^2-4r1r2=(r1-r2)^2>=0. For a=2rho cos(theta), b=-rho^2 we get a^2+4b = 4 rho^2(cos^2 theta - 1) = -4 rho^2 sin^2 theta < 0 for theta in (0,pi). Equality Delta=0 forces r1=r2 (real) or sin theta=0 (excluded). Since (a,b) come from three samples, Delta does too, with no dependence on L. QED

Boundary. Requires exact h(0..2); with noise the discriminant is a fragile estimate (NT08 gives the robust version). Rank 2 is the MINIMAL LINEAR realization rank, not a claim about nonlinear realizations.

Architectural use. An O(1) architecture diagnostic: before training, three impulse samples tell whether a grid is real-pole or conjugate-pole and whether its roots coincide (collinearity risk, NT06). Useful for init_mode validation in sgc.py. *Source:* core/arch/sgc.py _init_oscillator_grid / build_kernel

A.18 NT18 - Linear grids maximize the minimum spacing under a theta_mi

Premises. M frequencies theta_1 < ... < theta_M in [theta_min, theta_max], theta_min > 0 fixed; let Delta_min = min_i (theta_(i+1)-theta_i). A grid is admissible if it respects the interval.

Claim. (a) Delta_min <= (theta_max - theta_min)/(M-1), with EQUALITY iff the grid is exactly equally spaced (linear). (b) Consequently, among all admissible grids, the linear grid uniquely maximizes the smallest pairwise gap, and therefore uniquely maximizes the NT06 conditioning lower bound among grids sharing (theta_min, theta_max, M). (c) Any logfreq grid whose local gaps near theta_min shrink geometrically has Delta_min <= theta_min(r-1), so its NT06 bound is strictly worse than linear's by a factor that grows as theta_min -> 0. A hybrid grid is a convex interpolation and is bounded between them.

(a) Summing the M-1 gaps gives theta_M - theta_1 <= theta_max - theta_min. Hence (M-1) Delta_min <= sum gaps <= theta_max - theta_min, giving the bound. Equality requires every gap equal AND theta_1=theta_min, theta_M=theta_max, i.e. the linear grid.

(b) NT06 gives cond >= 2 sin^2(theta_1)/(Delta^2 A(rho)); with theta_1 and A(rho) fixed by the budget, maximizing the bound means maximizing Delta_min, which (a) shows is the linear grid, uniquely.

(c) For a logfreq grid theta_(i+1)=r theta_i, the smallest gap is theta_1(r-1) <= theta_min(r-1) (equality at the first gap). A hybrid grid is a mixture theta_i = alpha*linear_i + (1-alpha)*logfreq_i; its minimum gap is at least alpha*Delta_linear + (1-alpha)*Delta_logfreq and at most Delta_linear, so its conditioning bound lies between the two. QED

Boundary. Optimizes the MINIMUM gap / the NT06 LOWER bound, not the exact condition number (which is a measured value). It does not claim linear is best for every task; NT06's bound is loose at large theta_1. Theta ranges and M are fixed by the budget.

Architectural use. Turns 'linear30 was best in a 3-boundary sweep' from a measurement into a uniqueness theorem, and explains quantitatively why hybrid sits between legacy and linear (docs/frequency_grid_fix.md). *Source:* docs/frequency_grid_fix.md + core/arch/sgc.py init_mode

A.19 NT19 - The two-tap state is a MINIMAL sufficient statistic (tight byte bound)

Premises. Stable second-order IIR with distinct OR repeated roots, per basis function m; D channels; M basis functions; streaming state stores the previous two outputs per (m,d).

Claim. (a) No deterministic state of fewer than 2 scalars per (m,d) can reproduce the same output sequence for all inputs. (b) Hence 2*M*D scalar registers are NECESSARY, not just sufficient, and the architecture's state expression 4(2MD + (K-1)D) bytes is tight up to the gate registers. (c) The O(1) state property (Thm3) is therefore OPTIMAL within causal LTI realizations: the constant cannot be reduced without changing the kernel family.

(a) Suppose a 1-scalar causal linear state s_t = lambda s_(t-1)+u_t realizes the operator. Its impulse response is h(tau)=lambda^tau, so h(1)^2 = h(0)h(2). But NT17 gives h(0)h(2)-h(1)^2 = -b != 0 for BOTH root families (b=-r1r2 or b=-rho^2). Contradiction. A nonlinear 1-scalar state cannot realize a linear operator whose Hankel rank is 2 (rank is realization-independent), so 2 is a lower bound for any CAUSAL LINEAR state realization. (Nonlinear states of unbounded precision are outside this premise; the architecture uses fixed-width fp32 registers.) The recurrence itself achieves 2, so the minimum is exactly 2.

(b) M basis functions and D channels are independent (block diagonal in (m,d)), so the minima add: 2MD. Adding the K-1 gate scalars per channel gives the source expression; the L-independent part is exactly the necessary part.

(c) Any causal LTI realization of a degree-2 rational transfer function has McMillan degree 2; by (a) this is a hard lower bound, so no re-parameterization reduces the constant. QED

Boundary. Minimality is within CAUSAL LINEAR state realizations of the SAME kernel (the family used by sgc.py); unbounded-precision nonlinear registers are outside the premise. Changing the kernel family (e.g. first-order rho-only) changes the premise. Gate registers are architecture-specific overhead, not part of the tight bound.

Architectural use. Answers the reviewer's 'state constant is large' concern with a tight lower bound: the 56,320 B TSC state is a DIFFERENT (tensor) construction, while the original mixer's 512 B cannot be shrunk without losing the second-order kernel. *Source:* core/arch/sgc.py streaming_state_bytes + Thm3 + NT17

A.20 NT20 - Axiomatic uniqueness of the no-regression min-rule gate

Premises. A consolidation gate maps m per-metric paired-bootstrap CI lower bounds (ell_1..ell_m) and a tolerance tau to (accept,reject). Axioms: (P1) monotone: raising any ell_i cannot turn accept into reject; (P2) no compensation: if some ell_i < -|tau| then reject regardless of the others; (P3) permutation invariance; (P4) threshold invariance: the decision depends only on the bits a_i = 1[ell_i >= -|tau|]; (P5) non-triviality: a_i = 1 for all i implies accept.

Claim. The UNIQUE rule satisfying P1-P5 is G = AND_i [ell_i >= -|tau|], equivalently min_i ell_i >= -|tau|, which is exactly the upstream regression_gate. Consequently ANY weighted-sum, averaged, or majority-vote gate violates at least one axiom and permits a metric to collapse while still being compensated -- contradicting the source's 'no-regression' semantics.

By P4 the rule is a Boolean function g(a_1..a_m). By P2, any vector with some a_i=0 maps to reject. By P5 the all-ones vector maps to accept. These two facts determine g on all 2^m inputs: g = AND_i a_i. P1 and P3 are then automatically satisfied (AND is monotone and symmetric), so the rule exists and is unique. For a weighted sum S=sum_i c_i ell_i with c_i>0, choose ell_1 = -|tau|-epsilon and ell_2 = ... = ell_m = M large enough that S >= -|tau|: P2 is violated (a metric regressed but the round is accepted). Averaging and majority voting are special cases of such compensating rules. Hence they are excluded. QED

Boundary. Axioms P1-P5 are the formalization of the upstream docstring's semantics ('any metric fails -> reject'); the theorem is conditional on accepting that formalization. It concerns the AGGREGATION rule, not the statistical validity of the CI itself (that is NT15(a)).

Architectural use. Upgrades the gate from 'an engineering choice' to a forced characterization, and proves that the upstream implementation is the only one consistent with its own stated semantics. *Source:* simplcore/core/memory/dream/evaluation.py regression_gate

A.21 NT21 - The constant-verifier trap and necessity of an independent eva

Premises. The consolidation loop uses a verifier V(model) returning per-sample scores; the gate compares paired bootstrap CI lower bounds of new minus baseline scores; the loop refuses weight writeback when the verifier is the upstream DEFAULT (a constant function).

Claim. (a) If V is constant, every paired difference is exactly 0, every bootstrap resample has mean 0, the CI lower bound is 0 for every metric, and the gate therefore accepts EVERY round unconditionally: the gate carries ZERO information. (b) Hence 'refuse writeback when the default verifier is used' is not a conservative heuristic but a LOGICAL NECESSITY: it is the only way to keep the no-regression claim non-vacuous. (c) A non-constant, held-out verifier is NECESSARY and SUFFICIENT for the gate to be non-trivial (it can then reject).

(a) V(model)=c for all models implies score_new_i - score_base_i = c-c = 0 for every sample i. Any bootstrap resample of zeros has mean 0, so both one-sided bounds are 0 and 0 >= -|tau| holds; by NT20 the gate accepts. (b) If writeback were allowed in this case, the loop would commit arbitrary weights while claiming 'no regression proved', which is false because the proof is vacuous. Rejecting is the unique action that prevents a false claim. (c) Necessity: (a) shows a constant verifier gives a trivial gate. Sufficiency: if V is non-constant, there exist models with a negative paired mean; with enough samples the CI lower bound falls below -|tau| and the gate rejects (NT15(a)), so the gate is non-trivial. The upstream hooks.py default_verifier returns exactly 0.0 and loop.py:335 checks verifier_is_default before writing. QED

Boundary. Sufficiency in (c) is conditional on bootstrap consistency and sufficient sample size (NT15(a) premises). The theorem says nothing about whether any particular real verifier is well specified.

Architectural use. Formalizes a source-level guard that was previously only a comment, and closes the 'RSI gate is decorative' objection with a proof. *Source:* dream/hooks.py default_verifier; dream/loop.py VERIFY segment

A.22 NT22 - Irreducible gap of hidden-state alignment objectives (TS2SC)

Premises. Teacher is a content-addressed attention block; student is a stack of L static-LTI SGC mixers with a LINEAR residual stream (per-position FFN does not mix history tokens with the query before the final read-out); the alignment objective minimizes the residual-stream cosine to the teacher layer; content-addressing task as in NT13.

Claim. There exists epsilon_0 > 0, INDEPENDENT of L and of the student parameters, such that the alignment objective cannot be driven to zero: on some inputs the achievable residual cosine is at most 1 - epsilon_0. Adding layers does NOT close the gap, because each mixer preserves the premise that history enters only through a fixed linear summary. Hence 'fit the attention internals' is not a tuning problem but a function-class obstruction.

By NT13, a single static-LTI mixer with a linear residual stream cannot implement content addressing: there exist two value assignments v, v+Delta-v giving the same history summary A and the same query q, hence the same student output for any read-out, while the teacher's attention output differs by at least delta = ||E(v_(j1))-E(v_(j2))|| > 0. Therefore the student-teacher output discrepancy is at least delta on one of the two inputs for every parameter setting; this gives epsilon_0 > 0 after normalizing by the bounded residual norms. Stacking: the residual stream of L layers is a sum of layer outputs, each a static-LTI function of the same history plus per-position terms; under the linear-stream premise their SUM is again a fixed linear summary of the history (linearity of the sum), so NT13 still applies verbatim to the final stream. Consequently epsilon_0 does not depend on L. The measured ceiling 0.2484 vs oracle 0.2476 is a finite-sample ESTIMATE of this gap, not a theorem. QED

Boundary. Requires a LINEAR residual stream; per-position nonlinearities that mix A and q before the final read-out leave the premise (this is exactly conjecture C3, unproved). Does not forbid OUTPUT-distribution distillation (NT23), nor does it assert any specific numeric cos.

Architectural use. The formal reason the TS2SC weight-transfer study

could not succeed and why the 0.248 ceiling was 'eaten full', not 'one more coef tuning away'. It converts a project-level failure into a theorem. *Source:* TS2SC transfer study + NT13 + O1

A.23 NT23 - Separation of output-distribution distillation from hidden-state alignment

Premises. CONDITIONAL. (H1) The student's logit function class F_S is dense, in KL divergence, in the teacher's output distribution p_T on its support; (H2) the hidden-state alignment objective is as in NT22; teacher is frozen, distillation uses per-position KL with T>=1.

Claim. Under H1, inf over student parameters of KL(p_T || p_S) = 0, while the hidden-state alignment objective has the strictly positive irreducible gap epsilon_0 of NT22. Hence the two objectives are NOT equivalent: output distillation is not blocked by NT13/NT22, whereas internal alignment is. This holds WITHOUT assuming the student's internal states resemble the teacher's.

H1 states that for every eta>0 there exists f in F_S with KL(p_T||softmax(f/T)) < eta; taking eta->0 gives infimum 0. By NT22 the internal-alignment gap is >= epsilon_0 > 0 for all parameters, so its infimum is >= epsilon_0 > 0 = inf KL. Since the two infima differ, the objectives cannot be equivalent: no parameter sequence drives both to zero, and optimizing internal alignment cannot be justified as a surrogate for output KL. The proof uses only H1 and NT22; no claim about the student's hidden states is made. QED

Boundary. CONDITIONAL on H1 (density), which is NOT proved here and may fail for a narrow student logit class; the theorem is a SEPARATION, not a guarantee that KL reaches 0 in finite training. It does not claim output distillation succeeds empirically (O3 is a 3.07M-token pilot).

Architectural use. Gives the migration protocol a principled target: align the teacher's OUTPUT distribution, never its internals. Explains why the earlier ' + ' was the wrong objective. *Source:* TRANSFER_DISTILL_REPORT.md + NT22

A.24 NT24 - Hard stagnation of zero-order self-evolution (minimal-round bound)

Premises. NT15 holds: at most w insights of b bits are committed per round, T rounds, so I(history; weight increments) <= T*w*b bits. A task requires the weights to encode a sufficient statistic carrying I_task bits about the L-token history; the final read-out depends on the weights and the current query only.

Claim. (a) If I_task > T*w*b, the weights CANNOT encode a sufficient statistic for the task, so no read-out attains the task's Bayes error: self-evolution is HARD-STAGNATED before round T. (b) The minimum number of rounds satisfies T_min >= ceil(I_task/(w*b)), INDEPENDENT of L and of the bank capacity C. (c) Longer context does not shorten T_min; it only increases the retrievable history at inference, consistent with NT15(b).

(a) Let H be the history, Theta_T the weights after T rounds, and Y the task answer. If the weights encoded a sufficient statistic for Y then I(H;Theta_T) >= I(H;Y) >= I_task. But NT15(b) gives I(H;Theta_T) <= T*w*b < I_task, contradiction. Hence the Markov chain H -> Theta_T -> Y has bottleneck capacity below I_task and cannot achieve the Bayes-optimal error (data processing). (b) Rearranging T*w*b >= I_task gives T >= I_task/(w*b), and T is an integer. (c) Neither L nor C appears in the bound; by NT14(ii) only k selected items influence a read, so the unselected L-k items do not enter the rate. QED

Boundary. An INFORMATION-THEORETIC necessary bound, not a training-convergence claim and not an assertion that the loop reaches the rate. I_task is a property of the TASK, which must be estimated, not measured by accuracy. Does not forbid few-shot in-context retrieval (that uses the episodic bank, NT14).

Architectural use. Answers 'can long-context self-evolution keep improving?' with a provable NO beyond T_min, and converts RSI enthusiasm into a checkable budget inequality. *Source:* CT7(b) / NT15 + NT14(ii)

A.25 NT25 - Tightness of the quantization-resolvable radius

Premises. CONDITIONAL on a uniform-step absolute quantizer with step s, input bound |u_t| <= U, r = s/(2U) < 1, first-order state s_t = rho s_(t-1) + u_t with 0<rho<1; a contribution at delay tau is resolvable iff rho^tau U > s/2.

Claim. (a) The resolvable radius is exactly tau* = ln(r)/|ln rho| (floor), and it is TIGHT: there exists an input with a single unit pulse such that the contribution at tau*+1 is exactly at the quantization threshold and cannot be resolved, while all delays <= tau* are resolvable. (b) tau* = Theta((m+1)/|ln rho|) for m-bit mantissas, so the asymptotic rate is OPTIMAL up to a constant: no deterministic rescaling improves the EXPONENT. (c) For rho = 1-epsilon, tau* ~ (m+1) ln 2 / epsilon, giving the observed order of magnitude of the bf16/int8 long-length drop.

(a) rho^tau U > s/2 iff tau ln rho > ln(s/(2U)) = ln r, and since ln rho<0 this is tau < ln r / ln rho = ln(r)/|ln rho|. The largest integer below it is tau*. Choose the input with u_0=U and all other u_t=0: the state at delay tau is exactly rho^tau U, so at tau*+1 it equals or falls below s/2 by construction, hence is not resolvable; for every tau<=tau* it strictly exceeds s/2. No scaling of U or s changes r's logarithm by more than an additive constant.

(b) With s/U ~ 2^(-(m+1)), ln(1/r) = (m+1) ln 2 + O(1), so tau* = ((m+1) ln 2 + O(1))/|ln rho|. The dependence on |ln rho| in the DENOMINATOR is forced by the exponential decay law, so no deterministic rescaling alters the exponent; it can only change the O(1) numerator.

(c) |ln(1-epsilon)| = epsilon + O(epsilon^2), giving tau* ~ (m+1) ln 2 / epsilon. QED

Boundary. CONDITIONAL on a uniform absolute quantizer; the physics probe uses dynamic per-step scaling, so MEASURED drop points (L=2049/4097) are NOT derived from this theorem -- only the order of magnitude is. Tightness is within this model, not for all quantization schemes.

Architectural use. Upgrades the earlier 'conservative sufficient radius' into a TIGHT radius with an explicit precision-vs-decay rate, and explains NT07's interior optimum quantitatively. *Source:* Appendix A.7 perturbation bound + O6 physics probes + NT07

A.26 NT26 - Causal optimality of per-step peak calibration

Premises. CONDITIONAL. Streaming state s_t, causal information set at step t contains only I_t = (|u_s| <= U, s_(t-1), rho), i.e. no future peak is known. A quantizer chooses a scale Delta_t >= |s_t| to avoid overflow and incurs absolute error delta_t proportional to Delta_t. A 'fixed global scale' uses Delta_t = Delta for all t.

Claim. (a) The per-step peak rule Delta_t = max(|s_t|, Delta_min) is the POINTWISE MINIMAL admissible scale given I_t, hence minimizes delta_t at EVERY step. (b) Any fixed global scale must satisfy Delta >= sup_(t<=L)|s_t|, and since sup >= max_t|s_t| pointwise, delta_fixed >= max_t delta_peak(t), with strict inequality whenever the peak varies. (c) Therefore fixed-scale quantization is never better and is strictly worse on any sequence with a non-constant state envelope; the improvement is CAUSAL (needs no future information).

(a) Admissibility requires Delta_t >= |s_t|; the smallest such value is |s_t| (or Delta_min for the zero state), which uses only s_(t-1), u_t and rho -- all in I_t. Minimality is pointwise, so delta_t = c Delta_t is minimized pointwise.

(b) A fixed Delta must cover every t<=L, so Delta >= sup_t |s_t| >= |s_t| = Delta_peak(t) for each t; hence delta_fixed >= delta_peak(t) for all t, i.e. >= max_t delta_peak(t). If the envelope is non-constant the inequality is strict at the step where |s_t| < sup.

(c) Combining (a) and (b) gives the partial order, and (a) shows the rule is implementable causally. QED

Boundary. CONDITIONAL on scale-proportional absolute error and the stated causal information set. It proves a PARTIAL ORDER of error bounds, NOT that dynamic scaling always wins empirically (O6's fp16 non-monotonicity is a separate genuine numerical effect). It does not contradict the withdrawal of 'dynamic scaling causes spurious artifacts' -- those are different claims.

Architectural use. Upgrades the previously DEMOTED one-line partial order (T14) into a causal-optimality theorem, and is exactly the standard that separates 'too ordinary to write' from 'non-trivial enough'. *Source:* O6 fixed-scale controls + T14 + docs physics frontier

A.27 NT27 - Phase transition of the optimal decay across precision depths

Premises. CONDITIONAL. NT07's resolvable-length bound g(rho) = ln((1-rho)U/(c delta)) / |ln rho| on rho in (0, hi), hi = 1 - c delta/U; delta/U = 2^(-(m+1)) for mantissa depth m; c>1 fixed; delta/U sufficiently small.

Claim. (a) g is strictly increasing near rho -> 0 and strictly decreasing near hi, so for small delta/U it has a SINGLE interior maximum rho*(m). (b) rho*(m) solves the stationarity equation (1-rho) ln(1/rho) ln((1-rho)U/(c delta)) = rho ln(1/rho) + ... equivalently the balance (1-rho) ln((1-rho)U/(c delta)) = rho ln(1/rho); asymptotically rho*(m) = 1 - Theta(2^(-m)). (c) Therefore DIFFERENT precision depths have DIFFERENT optimal decays: a single rho tuned for fp32 is provably suboptimal for int8. rho -> 1 is never optimal (NT07).

Write N(rho)=ln((1-rho)U/(c delta)) > 0 and D(rho)=ln(1/rho)>0, g=N/D. Then g' = (N' D - N D')/D^2 with N' = -1/(1-rho) < 0 and D' = -1/rho < 0, so sign(g') = sign(-D/(1-rho) + N/rho). As rho -> 0, D = ln(1/rho) -> +infinity but N -> ln(U/(c delta)) is finite and positive; the term N/rho ~ ln(U/(c delta))/rho grows as 1/rho and therefore dominates -D/(1-rho) ~ -ln(1/rho). Hence g' > 0 near 0. As rho -> hi, N -> 0+ while D stays positive and finite; then N/rho -> 0 and -D/(1-rho) < 0 dominates, so g' < 0 near hi. By continuity g' has a zero in (0,hi); for small delta/U the ratio N/D is unimodal (one sign change), giving a UNIQUE interior maximum rho*. Setting g'=0 gives N/rho = D/(1-rho), i.e. (1-rho) N = rho D, and the stated balance. Now put rho = 1-epsilon with epsilon small and delta/U = 2^(-(m+1)): then D = ln(1/(1-epsilon)) = epsilon + O(epsilon^2) and N = ln(epsilon * 2^(m+1)/c) = (m+1) ln 2 + ln epsilon - ln c. The balance (1-rho)N = rho D becomes epsilon * N = (1-epsilon) * (epsilon + O(epsilon^2)), so for epsilon>0 we need N = 1 + O(epsilon). Substituting N gives ln epsilon = 1 - (m+1) ln 2 + ln c + O(epsilon), hence epsilon = c*e*2^(-(m+1)) (1+o(1)) = Theta(2^(-m)) and rho* = 1 - Theta(2^(-m)). This matches the NT07 finite scans (m=8 -> 1-rho* ~ 1e-2, m=11 -> ~1e-3, m=24 -> ~3e-8). Since rho* depends on m, a single rho cannot be optimal for two different mantissa depths. QED

Boundary. CONDITIONAL on NT07's uniform-error model; uniqueness of the stationary point is asserted for SMALL delta/U (the regime of interest); it is not a global proof for all parameters. The asymptotic rho*=1-Theta(2^(-m)) is analytic, and is CROSS-CHECKED against NT07's finite scans but not derived from them. Measured O6 rho=0.999 is an observation.

Architectural use. Explains why the int8 / bf16 / fp32 frontiers must be tuned separately and gives the precision-dependent decay law; strengthens NT07 from 'an interior optimum exists' to 'how it moves with m'. *Source:* NT07 + O6 physics_precision.json

A.28 NT28 - Layer x tensor-order incompatibility inequality for H-hop retrieval

Premises. A stack of L layers; each layer's output is a polynomial of total degree at most the tensor order o of its state contraction (order-2 outer product gives o=2; bilinear contraction degree 2). Skip connections do not raise degree. The H-hop target contains H key-match factors plus one value factor, total degree H+1.

Claim. L * ln(o) >= ln(H+1), equivalently o^L >= H+1. For o=2 this recovers NT11 (L >= log2(H+1)); for L=1 it forces o >= H+1, so a SINGLE layer needs tensor order at least H+1. CRITICALLY: increasing the number of pole channels P changes the KERNEL SHAPE but not the tensor order o, so P cannot compensate for too few layers or too low an order. The two resources (depth, order) are incomparable and must both satisfy the product bound.

Composition of polynomials of degrees d_1..d_L has degree at most prod d_i <= o^L; skip connections (addition) do not increase degree. The H-hop target is a multilinear form of degree H+1 (H factors <phi(q),psi(k)> plus one value). A polynomial of degree < H+1 cannot equal a nonzero degree-(H+1) form, so o^L >= H+1, giving the inequality. Setting o=2 gives NT11; setting L=1 gives o>=H+1. The pole count P enters only through K_n(d)=sum_p w_(p,n) rho_(p,n)^d, a SCALAR multiplier of each delay; it does not add a new tensor index or raise the polynomial degree in the history variables. Hence P is orthogonal to the bound. QED

Boundary. NECESSARY condition only; no sufficient construction. The degree bound is an upper bound on what L layers of order o can reach. Per-position nonlinearities of degree > o change the premise. The measured H=4 multihop=0.0322 is a finite-sample observation.

Architectural use. Unifies NT01 (order 2 for 1 hop) and NT11 (layers for H hops) into one inequality, and PROVES that the 'add more poles' fix cannot work -- closing a whole search direction that was tried implicitly. *Source:* NT01 + NT11 + O5 H4_multihop

A.29 NT29 - Topological necessity of a discontinuous hard pointer

Premises. Exact retrieval map R(q) = v_(j*(q)) with j*(q) = argmax_j <phi(q), k_j>, where >=2 distinct keys k_1,k_2 have distinct values v_1 != v_2 and the key space is a continuum containing the bisector hyperplane between them. The state is constant-size (no per-token cache).

Claim. (a) R is DISCONTINUOUS on the bisector, hence not differentiable there; hard pointer selection is therefore a NECESSARY feature of exact retrieval, not an implementation artifact. (b) Any CONTINUOUS surrogate Rtilde that agrees with R off an epsilon-neighborhood of the bisector has a NONZERO mis-selection probability bounded below by the measure of that neighborhood under any query distribution with positive density there. (c) Softening cannot simultaneously be continuous AND exact; this is an INDEPENDENT obstruction from NT14(iii)'s kernel-collision argument (which concerns the aggregate read-out, not the selector).

(a) Let q_1,q_2 be points on opposite sides of the bisector B = (q : <phi(q),k_1> = <phi(q),k_2>) with <phi(q_1),k_1> > <phi(q_1),k_2> and the reverse at q_2. Then R(q_1)=v_1 and R(q_2)=v_2. Take a path q(s) from q_1 to q_2 crossing B at s0; the left limit of R is v_1 and the right limit is v_2 != v_1, so R is discontinuous at s0 and has no derivative there.

(b) A continuous Rtilde must pass from v_1 to v_2 continuously, so on the epsilon-neighborhood N of B it takes values strictly between them; no point of N is mapped to the exact correct value v_1 or v_2. Hence P[Rtilde(q) != R(q)] >= P[q in N] > 0 for any query law with positive density on N.

(c) NT14(iii) shows a differentiable SOFT AGGREGATE over all slots has a kernel collision; (a)-(b) show the SELECTOR itself cannot be continuous. They are different maps with different proofs, so the two obstructions are independent. QED

Boundary. Requires a continuum query space and >=2 distinguishable keys; a finite/discrete query alphabet could in principle use a lookup table (but that is itself a discontinuous map on the embedding space). Does not forbid approximate retrieval (NT08).

Architectural use. Gives the L0/L1/L2 three-tier sidecar its theoretical justification: the L2 top-k POINTER must be hard/non-differentiable, which is why it sits outside the differentiable SGC mixer and cannot be folded into end-to-end training. *Source:* docs L0/L1/L2 tier design + memory.py topk + NT14

A.30 NT30 - End-to-end tightness chain: none of the five bounds can be imp

Premises. Exact H-hop retrieval over V token classes, N value dimensions, M second-order basis functions, in a static-LTI constant-state architecture with no N x N scoring and no growing KV cache; NT01, NT04, NT05, NT19, NT28 and NT13 all hold.

Claim. Any such architecture simultaneously requires (i) tensor order o >= H+1 for L=1, or o^L >= H+1 in general (NT28); (ii) code dimension R >= V (NT04); (iii) state >= 2*M*R*N scalars, with R < 2V attainable (NT05, NT19); (iv) a strictly positive kernel, hence real-pole roots (NT16/NT01). TSC at H=1 attains ALL FOUR within constant factors: order 2, R<2V, 2M*R*N<4M*V*N, real poles. Moreover the chain is JOINTLY TIGHT: improving any one bound alone cannot lower the product, because each is forced by a different independent theorem.

Chain the results: NT13 forbids order 1; NT01 achieves order 2 (tight). NT04(a) gives R>=V and NT04(b) gives R<2V (2-optimal, tight within the class). NT19 gives 2M scalars per code coordinate per value channel as a MINIMAL sufficient statistic, so with R code rows and N value channels the state is >= 2MRN; TSC uses exactly P*R*N with P=M poles, i.e. M*R*N first-order registers, and the second-order version 2MRN is attained by the recurrence (tight). NT16 forces strict positivity, hence real poles by NT02. For H hops NT28 forces o^L >= H+1, which no change in P, R, M or N can relax. Each inequality comes from a DIFFERENT proof (rank-nullity / orthogonality / Hankel rank / Turan / degree composition),

so relaxing one would contradict its own theorem without affecting the others: the product bound is jointly tight. TSC at H=1 meets all four simultaneously, so the bounds are not merely lower bounds but ACHIEVED ones. QED

Boundary. Tightness is up to constant factors (the factor 2 in R<2V, and the pole count P=M convention). It does NOT assert that no architecturally DIFFERENT construction (e.g. a non-LTI state) could beat the product; that would leave the premise. Neither LM parity nor throughput is implied.

Architectural use. The paper's single strongest summary statement: instead of one accuracy number, an end-to-end achieved tight bound. It also PROVES the project cannot be improved by 'add more poles / bigger grid / better code' alone -- only by leaving the premise (NT28). *Source:* NT01/NT04/NT05/NT16/NT19/NT28 consolidated

Appendix B Conjectures and Open Questions

Explicitly unverified. None is a premise of any theorem or a claimed result. NT04 closes C2; NT11 gives C1/C3 a necessary condition but no construction.

· **C1.** Stacking TSC layers or inter-layer recurrence restores multi-hop retrieval. NT11 gives the layer lower bound; no stacking experiment exists.

· **C2.** CLOSED by NT04: a deterministic orthogonal code exists for every V with R < 2V.

· **C3.** NT13 extends to deep stacks with per-position nonlinearities. Cross-terms between A and q leave the fixed-summary premise; unproved.

· **C4.** Freezing rho = 0.999 from the start of training remains stable on >=10M-token language modeling. Verified only on a synthetic task.

· **C5.** Layered quantization or residual accumulation pushes the int8 long-length boundary further. NT07 bounds the trade-off; untested schemes remain.

· **C6.** On equal compute the periodic and content-addressing axes are complementary. The periodic-axis gain was NOT replicated (O4).

· **C7.** CLOSED IN PART by NT03: the rho<1 multi-pole kernel is provably not the linear-attention constant kernel. Full-model separation from all gated variants remains open.

· **C8.** The decay spectrum gives an inductive bias on oscillatory/damped physical signals that attention lacks. No measured comparison exists.

Appendix C Falsified Statements

· **F1.** Increasing rho_max extends the effective memory window. Rejected by measurement.

· **F2.** State-feedback rho_t/theta_t modulation (SFSC) preserves FFT duality. Rejected: LTV is non-Toeplitz (NT12); FFT-proxy relative error 0.9720.

· **F3.** Phase-resonance slots exceed equal-parameter attention. Rejected (O2).

· **F4.** Content addressing and FFT/IIR duality are mutually exclusive. WITHDRAWN via NT01/NT10; this withdrawal is a core contribution.

· **F5.** An oscillatory basis itself provides the inductive advantage. Rejected: a random orthogonal basis reached 1.0000; NT02/NT08 explain why.

· **F6.** Dynamic quantization scaling causes spurious non-monotonicity. Rejected by same-sample prefix controls.

· **F7.** Shannon capacity limits long-length retrieval here. Rejected: 176x headroom; NT07 identifies the decay/precision trade-off instead.

· **F8.** A monotone verifier score implies monotone self-improvement (former T16). WITHDRAWN (W7); trivial scalar induction and inconsistent with the multi-metric paired-bootstrap gate. Replaced by NT15.

Appendix D Observation Provenance

All values were computed from raw result files accompanying the anonymized supplementary artifact; none was transcribed from a secondary summary.

Obs.	Config / seeds	Key values	File
O1	1.7B transfer, 1 run	0.2484 / 0.2476 / 0.5430	transfer study
O2	single-hop, 3 seeds	0.8672 vs 1.0000	phase-resonance probe
O3	470K, 3.07M tokens	10.817 / 11.281 / 10.217	native pilot
O4	7 grids, 1 run	0.1094-0.1354 vs 0.1575	periodic-axis probe
O5	V=64, 5 seeds, 512/seed	1.0000 / 0.9987 / p=0.4998	TSC probe
O6	5 seeds, 4 precisions	L=16385, 43/43 checks	physics probes

Measured configuration details. O5: W1 has 10264 parameters, W4_half 4920, M5 2792; the stored reference record has 10240. **Fisher p = 0.4998 means no significant difference was detected, NOT equivalence.** The paired bootstrap on the general second-order dictionary gave mean difference 0.1901 with 95% CI [0.1582, 0.2220] over 3 seeds (sample-internal variance only). O6: frozen-rho retraining held-out means [1.0, 1.0, 1.0, 1.0], between-seed SDs [0.0, 0.0, 0.0, 0.0]; L = 16,385 accuracy mean 1.0000, min 1.0000; per-token cost at L = 4097 was 35.25 us with a 56320 B state. Conservation guards reported FFT/IIR relative errors of order 1e-16.

Appendix E Reproducibility and Tool Use

Every theorem is derived analytically in Appendix A and requires no execution. Every observation comes from a deterministic script with recorded seeds, precision and sample counts. The supplementary artifact contains raw result files, gradient finite-difference checks, conservation assertions, the 49-item audit (49/49 PASS) and the theorem ledger. Gradient checks passed 8/8, 10/10, 28/28 and 44/44; the physical consistency checker passed 43/43. No training run is used to support any theorem.

Generative AI tools were used for implementation assistance, experiment scripting and language editing. They did not originate the research questions, conjectures or conclusions; no experimental data, citation or result was generated by such a tool. The human author verified every numerical claim against raw files and accepts full responsibility.

Appendix F Withdrawn Claims Register

Earlier working-version statements, formally withdrawn so the revision is auditable. Withdrawals are contributions, not hidden errata.

Withdrawn statement	Reason
Unconditional fp32 bitwise duality	Different rounding orders; real-field only (NT09)
Content addressing <-> duality exclusive	WITHDRAWN, core contribution: NT01/NT10 counterexample
Periodic-axis superiority over attention	Not replicated in-repository (O4)
Equivalent 4-7B parameter claim	No supporting measurement
NT13 holds for any read-out unrestricted	Exactly four premises required; stated in abstract
Dynamic scaling causes spurious effects	Same-sample prefix controls (O6)
W7: monotone score => self-improvement (T16)	Trivial induction + mismatched source gate; replaced by NT15
F8: scalar monotone score as gate	NT20 proves the gate is the unique multi-metric AND rule
W8: content addressing needs input-dependent coefficients	WITHDRAWN, second core contribution: NT31 attains H-hop retrieval with STRICTLY static kernels
W9: absolute-position parity is a general role basis	WITHDRAWN: hidden format prior; 1.0000 -> 0.3438+/-0.0314 under arbitrary spacing. Replaced by a 1-bit role tag (Lemma R1).

Appendix G The 51-Item Audit Ledger

Every statement in this project is adjudicated exactly once. The 31 theorem items plus ONE supporting lemma (R1) have complete analytic proofs in Appendix A; their finite-identity checks only certify correct formation and non-empty premises. The 19 non-theorem items are CLASSIFIED with a reproducible witness or an explicit falsification / experiment specification; PASS for those means 'correctly classified', NOT 'true'. No measured value is a theorem.

Class	Count	Why it is NOT a theorem (if applicable)
New theorems NT01-NT31	31	NT31 H5 revised by W9; 6 promoted to core
Supporting lemma R1	1	Analytic, non-trivial; deliberately NOT a headline theorem
Falsified F1-F8	8	Explicit counterexamples / witness values
Withdrawn W1-W5, W7	5	Formally retracted earlier working claims
Conjectures C1, C3, C7	3	Unproved; falsification experiment specified
Empirical E1-E3	3	LM parity / SOTA / throughput: no theorem can establish

G.1 The 30 theorems (six de-duplicated groups)

· **I: TSC construction and optimality.** NT01, NT02, NT03, NT04, NT05

· **II: grid, precision and length.** NT06, NT07, NT08, NT09, NT10

· **III: multi-hop, self-evolution, equivalence.** NT11, NT12, NT13, NT14, NT15

· **IV: algebraic structure and identifiability.** NT16, NT17, NT18, NT19

· **V: self-evolution and transfer objectives.** NT20, NT21, NT22, NT23, NT24

· **VI: numerical frontier and multi-hop duality.** NT25, NT26, NT27, NT28, NT29, NT30

G.2 The 19 non-theorem statements

· **F1 [FALSIFIED].** Increasing rho_max extends the window; NT07/NT27 show the optimum is interior and rho->1 shortens resolvable length.

· **F2 [FALSIFIED].** SFSC time-varying rho_t/theta_t preserves FFT duality; time-varying a_t gives non-Toeplitz h (NT12), FFT-proxy error 0.9720.

· **F3 [FALSIFIED].** Phase-resonance slots exceed equal-parameter attention; witness 0.8672 < 1.0000 (5,217 vs 5,152 params).

· **F4 [FALSIFIED].** Content addressing <-> FFT duality mutually exclusive; withdrawn via NT01/NT10. This withdrawal IS a core contribution.

· **F5 [FALSIFIED].** The oscillatory basis itself gives the advantage; random orthogonal code also reaches 1.0000 (NT08), and NT16 shows conjugate poles are not strictly positive.

· **F6 [FALSIFIED].** Dynamic scaling causes spurious artifacts; same-sample prefix controls, and NT26 proves per-step peak is causally pointwise-minimal.

· **F7 [FALSIFIED].** Shannon capacity limits long-length retrieval; 1,280 required bits vs 225,280-bit state (176x headroom), noise 0.20 leaves accuracy unchanged.

· **F8 [FALSIFIED].** A scalar monotone verifier score is the gate; NT20 proves the gate is the UNIQUE multi-metric AND rule.

· **W1 [WITHDRAWN].** Unconditional fp32 bitwise duality; real-field only (NT09).

· **W2 [WITHDRAWN].** Periodic-axis superiority over attention; not replicated (O4).

· **W3 [WITHDRAWN].** Equivalent 4-7B parameter claim; no supporting measurement.

· **W4 [WITHDRAWN].** NT13 holds for any read-out unrestricted; exactly four premises required.

· **W5 [WITHDRAWN].** Dynamic scaling causes spurious effects; same-sample prefix controls (O6).

· **W7 [WITHDRAWN].** Monotone score => self-improvement (T16); trivial induction and mismatched with the source gate; replaced by NT15.

· **C1 [CONJECTURE].** Stacking TSC layers restores multi-hop retrieval. NT11/NT28 give only NECESSARY lower bounds; falsify with L=2,3 stacks on H=2,3, >=5 seeds.

· **C3 [CONJECTURE].** NT13 extends to deep stacks with per-position nonlinearities; cross-terms leave the fixed-summary premise; falsify with a 2-layer FFN-mixed counterexample.

· **C7 [CONJECTURE].** TSC is outside EVERY gated linear-attention variant; NT03 settles only rho<1 multi-pole vs rho=1 single-pole; falsify by exhibiting a gated variant with the same kernel family.

· **E1 [EMPIRICAL].** >=10M-token equal-compute LM parity; requires >=10M tokens, >=5 seeds, paired CI, and same-run Mamba-2/GLA/DeltaNet/Hyena/TS baselines. O3 is a 3.07M-token pilot.

· **E2 [EMPIRICAL].** RSI reaches SOTA / general-language superiority; NT24 gives a necessary rate bound, not capability. No theorem can establish this.

· **E3 [EMPIRICAL].** Absolute throughput advantage over attention; O6 is an unoptimized reference implementation, only scaling is informative; requires a measured same-run comparison.

Why 51 statements is not the target. Eight statements are demonstrably false, five were formally retracted, three have no proof, and three are empirical claims about scale and capability. Promoting any of them would reintroduce exactly the theorem inflation and 'numerical verification posing as proof' that the withdrawn-claims register (Appendix F) exists to prevent. The competitive contribution is 6 core theorems, 31 audited proofs, 1 supporting lemma, plus an auditable account of the 19 statements that must NOT be theorems.